

UNIT –I

Introduction, Benefits and challenges, Cloud computing services, Resource Virtualization, Resource pooling sharing and provisioning

1. INTRODUCTION:

Cloud Computing : It is storing and accessing data application and resources over the internet “the cloud “ instead on your own computer or physical server, allowing for **on demand** and **flexible resource management**.

Cloud computing represent an advances in the field of computing technology where the management of computing services is outsourced to greater extents and in multiple dimensions.

Computing facilities are made available as services offered by reputed vendors.

The developments in the field of **computing hardware, software architecture, and web technology and network communications** over the last decade. The speed of internetworks has increased day by day and it has also become cheaper. All these developments contributed in setting the stage for the initiation of the revolutionary concept of ‘cloud computing’.

Cloud computing provides the means for users **to easily avail computing facilities** whenever and wherever required. They need not worry about setting up infrastructure, purchasing new equipment or investing in the procurement of licensed software. Rather they can access any volume, large or small, of computing facilities in exchange for some nominal payment.

It is a new model of computing which has become possible through integration of advanced computing model sophisticated web technologies and modern network communication.

1.2 LIMITATIONS OF THE TRADITIONAL COMPUTING APPROACHES

Every new technology emerges with the promise to resolve the shortcomings of the existing ones. Traditional computing has played a pivotal role in the field of computing and communication over the past few decades.

Limitations of the traditional computing approaches

- In previous years enterprises used to execute their businesses merely by the aid of the pen, paper, telephone and fax machine.
- Gradually, computer systems intruded manual processes and started automating them.
- The pen and paper were replaced by digital communication, and even the phone and faxing services started being managed by computers.
- Businesses from local to global, are dependent on the computing systems for almost everything they do.
- Even individuals depend heavily on computing systems for their day-to-day activities.
- Easy and cheap access to computing facilities has become essential for everyone.

PROBLEMS ASSOCIATED WITH TRADITIONAL COMPUTING APPROACHES:

1. **Enterprise Perspective**
2. **Individual User's Perspective**

1. **Enterprise Perspective:** Enterprises are the one who adopt computing based automation for running day to day business activities

Difficulties faced by enterprises in traditional computing

Traditional Computing Scenario	Problematic facts and related questions
Business without help of computing services is beyond imagination, and the customized software packages manage business activities. Most organizations use ERP packages (implemented by some IT enterprise) to get maximum benefits from regular business operations.	To run enterprise resource planning applications, business organizations need to invest huge volumes of capital to setup the required IT infrastructure. Servers, client terminals, network infrastructure are required, and they to be put together in a proper manner. Moreover, arranging adequate power supply, cooling system and provisioning space also consume a major part of the IT budget. Are there ways to avoid this huge initial investment for computing infrastructural setup?
Business application package implementation also over-burdens the IT enterprises with many other costs. Setting up infrastructure, installation of OS, device drivers, management of routers, firewalls, proxy servers etc. are all responsibilities of the enterprise in traditional computing approach.	Enterprises (or IT service firms) need to maintain a team of experts (system maintenance team) in order to manage the whole thing. This is a burden for HR management and incurs recurring capital investment (for salaries). Can enterprises get relief from these responsibilities and difficulties? It would help them concentrate fully on the functioning of business applications
Even those IT enterprises whose sole business interest is developing applications are bound to setup computing infrastructure before they start any development work	This is an extra burden for enterprises who are only interested in application development. They can outsource the management of infrastructure to some third party, but the cost and quality of such services varies quite a bit. Can IT enterprises avert such difficulties?
Computing infrastructure requires adequate hardware procurement. This procurement is costly, but it is not a one-time investment. After every few years, existing devices become outdated as more powerful devices appear	It becomes difficult to compete in the market with outdated hardware infrastructure. Advanced software applications also require upgraded hardware in order to maximize business output. Can this process of upgrading hardware on a regular basis be eliminated from an enterprise's responsibility?
It is not unusual to find an updated version of application with new releases that is more	Adopting an updated version of an application requires necessary efforts from subscriber's end.

advanced and apt to keep up with changing business scenario	Fresh installation and integration of components need to be done. Can subscribers be relieved of this difficulty of periodically upgrading the applications?
Capacity planning of computing resources is a critical task for any organization. Appropriate planning needs time, expertise and budgetary allocation since low resource volume hampers the pace of the performance of applications	Enterprises generally plan and procure to support the maximum business load that they have anticipated. But average resource demand remains far less, most of the time. This causes resource wastage and increases the recurring cost of business. If this capacity planning task could be made less critical and resource procurement strategy more cost effective?
Resource requirements of a system may increase or decrease from time to time	Individual enterprises cannot manage system contraction in a way that unutilized resources of a system can be utilized in some other system so that the cost of the business could be reduced. If this were somehow possible?
Many enterprise computing systems run forever without stopping. Such systems host applications which require round-the-clock availability to fulfill business demand	When resource capacity expansion of such system becomes an absolute requirement for the respective business, a system shutdown (hence service disruption) becomes unavoidable which may cause loss in the business. If a system could be expanded without shutting it down?

An enterprise can be overburdened when handling multiple issues in the traditional way of computing. Issues like huge initial investment for setting up computing infrastructure

- difficulty of managing the complex systems,
- difficulty in regularly replacing outdated things
- with latest available technology etc

2. Individual User's Perspective

Individual users have also been consumers of computing since a long time. They use computing for various purposes like editing documents, developing program, playing games, watching videos, accessing Internet etc.

Difficulties faced by individual users in traditional computing approach

Traditional Computing Scenario	Problematic facts and related questions
To work with software applications (like text editor, image editor, programming, games etc.), users first need to procure computing system where these software applications run.	For general users (who don't want to experiment with computer hardware devices), this initial capital investment for setting up computing infrastructure is often more than the software applications they use! If this huge unnecessary investment for procuring hardware components could be avoided?
Requirement analysis and procurement of hardware infrastructure are responsibility of the users. But, actual utilization of these resources depends on frequency of user access and the kind of software applications they run over it.	General users are usually not experts of computing systems. They are often misguided and procure unnecessary volume/capacity of hardware resources, most portions of which remain unutilized. This reduces the return on investment (ROI). If this approach could be changed; if users did not have to procure a fixed volume of hardware resource prior to its actual use or demand?
A hardware component may fail for many reasons. Maintenance of the hardware infrastructure is the users' responsibility.	Time, cost and the uncertainty are involved in the process. If users could get relief from these responsibilities and difficulties?
Computing systems (desktop, laptop etc.) procured by most users are hardly used for few hours daily on an average.	Non-utilization of procured systems results in wastage of resource with regard to total investment. If the hardware resources would be available on payment of usage basis?
Software licensing cost needs separate budgetary allocation. Licenses are sold for fixed period of time. (usually for one year-duration).	If software is used for 2–5 hour per day on an average during licensing period, it depicts 8%–20% utilization of entire investment. If this cost could be reduced? If the licensing fee would be paid on hourly usage basis?

Users are burdened with installation and critical customization of software. They also troubleshoot in case the software crashes.	Professional help can be obtained against payment, or users can troubleshoot themselves, thereby investing more time. If users could get relief from these responsibilities and difficulties?
Users need to have physical access to the system for using a personal computing system.	Though portable computing devices are available (like laptop, tablet etc.), it may not be possible to carry them all the time. Could there be a way of accessing personal computing systems remotely, from any location, any time?
Within few years, hardware systems become outdated. It becomes difficult to run advanced or new software on them.	Users have no other option but to throw out the whole setup and replace it with a new one. If there be a permanent solution to this wastage (from users' end)?

IS THERE ANY SOLUTION TO THESE

When an organization or an individual user needs to set up in-house IT infrastructure (small or big) for computing purpose, they have two options:

1. Doing it themselves (by deploying in-house team, in case of an organization)
2. Outsourcing the responsibility to some third party who are well-trained for the job

IT service outsourcing is a popular model which has been adopted by enterprises over the last two decades, where all of the application development and implementation related activities/ responsibilities are transferred to IT service companies. But, traditional outsourcing is not the best solution to all problems mentioned above.

For example, enterprises that only use applications to run their business activities have some kind of computing requirements, but those IT service organizations to which they outsource the complex tasks of application development or implementation have their separate computing requirements.

Application development team of an IT service organization may in turn depend on some system (physical computing system), whereas the assembling and maintenance group who need not to have Knowledge of application development may have concerns that are entirely different.

They will certainly have separate requirements. Difficulties faced by users (of computing) depend on the layers of computing they work on.

THREE LAYERS OF COMPUTING

Computers and computing have become an integral part of our daily lives. Different people use different categories of computing facilities. These computing facilities can be segmented into three categories:

1. Infrastructure
2. Platform
3. Application

These three categories of computing facilities form three layers in the basic architecture of computing. Figure 1.1 represents the relationships between these three entities.

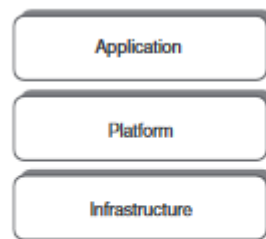


FIG 1.1: Three layers of computing facilities

1. Infrastructure

The bottom layer or the foundation is the ‘computing infrastructure’ facility. This includes all physical computing devices or hardware components like the processor, memory, network, storage devices and other hardware appliances. Infrastructure refers to computing resources in their bare-metal form (without any layer of software installed over them, not even the operating system). This layer needs basic amenities like electric supply, cooling system etc.

2. Platform

In computing, platform is the underlying system over which applications run. It can be said that the platform consists of the physical computing device (hardware) loaded with layer(s) of software where the program or application can run. The term ‘computing platform’ refers to different abstract levels. It consists of:

1. Certain hardware components, only.
2. Hardware loaded with an operating system (OS).
3. Hardware and OS, additionally, loaded with run-time libraries.

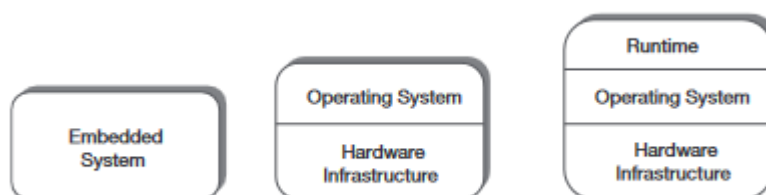


FIG 1.2: Computing platforms in different forms

Hardware alone can be considered as the platform in case of embedded systems, where physical resources can be accessed without any operating system

A fully configured physical computer loaded with an operating system is considered as a platform for computing. Different platforms can be installed over the same computing infrastructure. Linux or Windows operating systems installed over the same physical computer (computing infrastructure) can provide two different computing platforms.

The platform layer is also the place where software developers work. Hence Integrated Development Environments (IDEs) and runtimes are part of this layer. Java Development Kit (JDK) or .NET are examples of popular computing frameworks. Software applications can be developed and run over these platforms.

3. Application

Applications (application software) constitute the topmost layer of this layered architecture. This layer generally provides interfaces for interaction with external systems (human or machine) and is accessed by end users of computing. A user actually works on the application layer while he or she is going to edit a document, play a game or use the calculator in a computer. At this layer, organizations access enterprise applications using application interfaces to run their business.

Different types of people work at different layers of computing. They need to have different skill-sets and knowledge

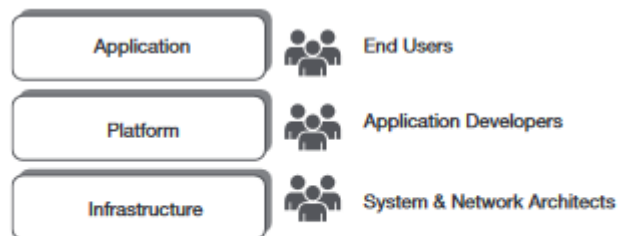


FIG 1.3: Different users/subscribers of three computing layers

Elements of these computing layers:

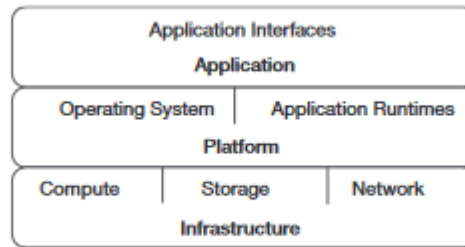


FIG 1.4: Elements of three computing layers

A person working online or offline with some software application on a personal computing device or directly accessing the Internet basically consumes all these three facilities together. The term ‘compute’ refers to the set of resources required for assembling a computing system or computer, particularly processor and memory components.

THREE LAYERS IN TRADITIONAL COMPUTING:

The boundaries between computing layers were not very clear to the general users.

Most end users unknowingly had to bear the burden of being concerned about all these three computing layers, while their actual area of interest was only around the application layer. Developers had to be concerned about the infrastructure layer apart from their own layer of activities (the platform layer).

Traditional Infrastructure Model:

Traditionally enterprises arrange **computing infrastructure themselves** by forming an internal computing system **management team** or they depend upon some **third-party vendors**.

These vendors or internal system management departments are responsible for managing the whole computing infrastructure of the enterprise, including

- Planning
- Designing
- Assembling of physical servers
- Network or data storage infrastructure build ups
- other services like load balancing of electricity supply.

The whole process of setting up of such an infrastructure is quite complex. For a new office, this process generally takes weeks and sometimes months.

Even the purchase cycle of additional hardware to power up the infrastructure is not counted in hours or minutes, but in days or weeks.

Protection and security of the infrastructure appears as an extra burden for any organization.

Traditional Platform Model:

Traditional model, the platform building and management tasks need special expertise. Enterprises either need to maintain an internal team or can outsource these jobs. Computing platform building activities involve installation of an operating system (OS), application runtimes or application development environments on computing infrastructure.

For example, Java Runtime Environment (which provides a platform for the applications) is essential to run any java based application.

Traditionally the entire responsibility of platform installation, configuration, updates and other staffing at appropriate levels often fall on the shoulders of the people concerned with application development or even on users who are interested only in using applications.

Thus, the actual assignments get delayed. Licensing, time bound installation of patches or maintenance of the platform cause difficulty for users (application developer or application user)

Traditional Application Model

At application layers, the users are the end users of computing. They need not have any knowledge about a complex computing system. Their only interests are to access or use different software applications to fulfill requirements of personal or business related works.

Software applications provide solutions for various needs. In traditional model, computing infrastructure or computing platform or both often become concerns of application subscribers, which is a critical scenario. Apart from this, the traditional software applications attract fixed and prepaid licensing costs and annual support costs.

The three main layers of computing, infrastructure, platform and application can be delivered to the consumers as ready-made stuff arranged and maintained by others, whenever they need it, relieving the consumers from the burden of arranging all the stuff themselves. This new model of computing is known as **cloud computing**.

Computing as the Utility Service

Cloud computing precipitated a significant shift in the responsibilities of managing computing resources.

Computing facilities in this model are supplied in the same way as like as a civic authority supplies water or electricity in a city.

Customers can use those facilities without being worried about how they are being supplied or who is managing all of these activities.

The three major aspects of computing which were represented in three-layered computing architecture.

Hence three services corresponding to this model are:

1. Infrastructure Service
2. Platform Service
3. Application or Software Service

The only thing required at customer's end to avail these readymade computing services(infrastructure, platform or software) are Internet connection and any basic computing device (PC, laptop, tab etc.) where

software interface with cloud computing systems can run. These computing devices need not be highly configured since the local computers no longer have to do heavy tasks and cloud interface applications are fairly light-weight.

Who Provides the Service?

Cloud computing vendors who are Independent Software Vendors (ISV) generally provide the **service**.

These vendors are reputed computing/IT giants. They are the owners or developers of the clouds and manage everything. Users can use cloud services by payment on use basis, just like they pay monthly bill for electricity service. This becomes profitable for both parties, the customers and the vendors. Vendors can supply the service at cheaper rates because of the large size of their business (due to, economy of scale) since the number of computing subscribers is very large.

This model of computing is much talked about because it provides a lot of flexibility compared to the traditional way of computing. For instance, it drastically reduces the cost and complexity of owning and operating computers and networks. Moreover, since specialized computing vendors manage the cloud the quality of service undoubtedly gets better.

The Concerns

Adoption of cloud computing also requires entrusting another company (the cloud vendor) with subscriber's personal, official and confidential data. But, these threats were all along existent in traditional computing where enterprises used to depend on IT service firms to, whom they outsourced various computing activities

Cloud computing promotes the idea of outsourcing of computing services, similar to the way electricity requirements are being outsourced today. Users can avail electricity without being bothered about its generation technique, where it comes from or how it is being transported. The only thing to know is that the usage is being metered and it will be billed as per the same.

PEOPLE ARE ALREADY USING CLOUD COMPUTING

Individuals as well as enterprises have been using cloud computing in some form or the other, through Internet.

When individuals use e-mail services like Yahoo Mail, Gmail or social networking services like facebook, twitter etc., they actually use a cloud computing service.

Picture or video sharing activities via mobile phones, which have been quite popular, are also based on the cloud computing model.

To access these services, users need not install any heavy application in local computing devices (like desktop, laptop, tablet etc.) apart from having some web browser or web-based app

INFLUENCES BEHIND CLOUD SERVICE ADOPTION

1. Technological Influences

2. Operational or Business Influences

The key technological influences behind cloud service adoption are

Universal Network Connectivity:

Cloud computing services are generally accessed through high speed network or Internet. Well-connected digital communication network spread across the world is necessary for ubiquitous access to cloud facility. As high speed network communication infrastructure has become available around the world, access to the cloud computing facility from any location has become a reality.

High-Performance Computing:

High-performance computing (HPC) systems needed specialized hardware components which were costly. Affording HPC was once beyond the imagination of small enterprises and individuals. Cloud computing has made HPC affordable for everyone by aggregating computing power to produce computing performance for executing high performance tasks.

Commoditization:

A product or service turns into a commodity when it becomes marketable and can be interchanged with another product of same type, which should also be available in the market. This is possible when products or services from multiple vendors provide more or less same value to customers, and customers have the option of replacing one product with another product of some other vendor. Cloud offerings from different providers create the same scenario. This commoditization of cloud services has developed irrefutable marketplace for cloud adoption.

There are many **Business Factors** associated for moving into the cloud environment

1. Low Cost Solution
2. Outsourcing
3. Speed or Responsiveness
4. Automation
5. Small Initial Investment
6. Less Maintenance Cost
7. Mobility
8. Flexibility

Low Cost Solution:

With cloud computing, consumers adopt the philosophy of pay-as-you-use.

This turns out more cost effective, since **consumers no more need to pay for excess capacity** which was very common in traditional computing.

Outsourcing

IT outsourcing is a common phenomenon among enterprises. Cloud computing utility services provide scope for using facilities entirely managed by other

Speed or Responsiveness

The time required to develop, configure, or launch new systems in cloud computing environment

is much lesser in comparison with the traditional one.

Automation

Automatic availability of systems, recovery from failure, load balancing or performance monitoring of systems and applications, and auto-maintenance of computing systems are several features offered by cloud environment that provide a lots of advantage

Small Initial Investment

Even adopting latest technology also meant considerable amount of investment for existing users. Cloud computing eliminates these barriers, as customers need to invest very small capital to start.

Less Maintenance Cost

Studies show that on an average about 70 to 80 percent of any organization's IT budgets goes to the operation and maintenance of existing systems and infrastructure. This cost has drastically reduced in the regime of the cloud computing.

Mobility

Cloud services can also grow or shrink as needed. It can expand or shrink with business that provides the higher cost-benefit ratio. When a business needs more computing/IT support they consume more cloud services, when they need less they consume less. Since payment is on usage basis, **this elasticity** of cloud services provides great flexibility for the consumers. Cloud computing has eliminated this problem and it has become easier for anyone to avail the state-of-the-art computing facilities. Businesses can exhibit more speed and agility. Agility enables a system or business to exhibit quick response to changing atmosphere or situation without losing momentum.

Flexibility

Cloud services have emerged as a lucrative option for individual users of computing as well as for most enterprises. But, along with lots of driving forces few resisting forces also exist that decide the level and speed of moving towards cloud services.

II BENEFITS AND CHALLENGES

Cloud computing has introduced a real paradigm shift in the scope of computing. Unlike the conventional uses of computer technology, it facilitates computing as a **utility service** which is **delivered on demand**. The **computing facility** is managed by **providers** and can be measured in usage **volume** or usage **time**.

All these features of cloud computing provide several benefit:

- 1. Less Acquisition/Purchase Cost**
- 2. Reduced Operational Cost**
- 3. Reduced System Management Responsibility**
- 4. Use-basis Payment Facility**
- 5. Unlimited Computing Power and Storage**

6. Quality of Service
7. Reliability
8. Continuous Availability
9. Locational Independence/Convenience of Access
10. High Resiliency
11. Quick Deployment
12. Automatic Software Updates
13. No License Procurement
14. Safety against Disaster
15. Environment Friendly

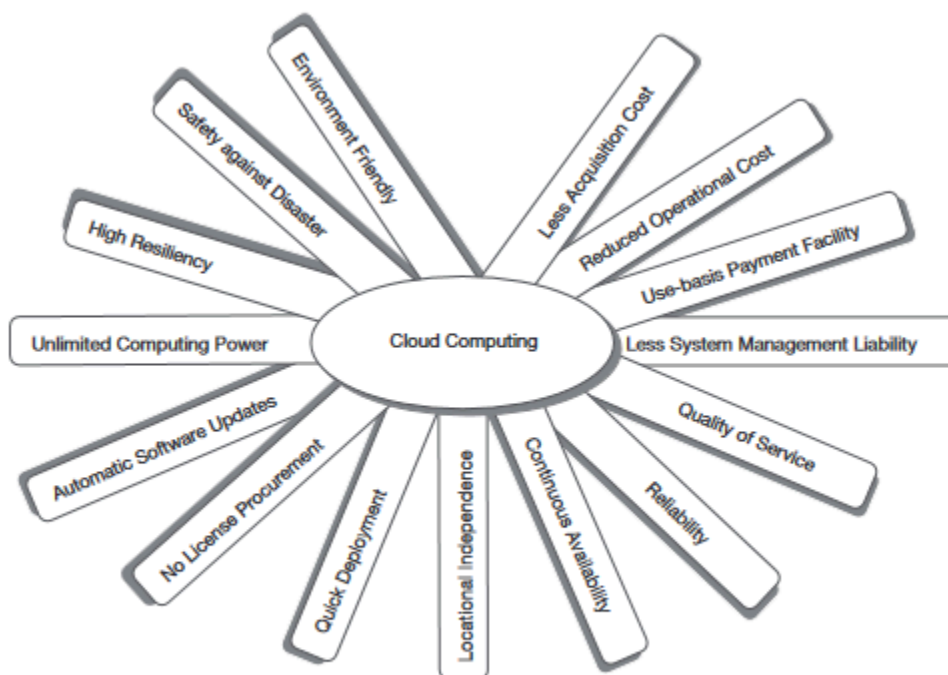


FIG 3.2: Advantages of cloud computing

1. Less Acquisition/Purchase Cost

- In **traditional computing**, users have to **purchase** or procure computing **resources** in significant amount at very beginning. Cloud computing is delivered following the utility service model.
- Since **vendor** arranges all necessary resources in this model, subscribers' **initial investment** for **acquiring hardware or software drops down drastically**.
- They need not to arrange anything apart from **client systems** to **access cloud services**. Thus, initial **capital expenditure** of user gets **reduced considerably**.

2. Reduced Operational Cost

- utility computing the **cost of running any systems round the clock moves towards the provider's end**. Subscribers get rid of the responsibility of system administration, maintenance, and 24 × 7 energy support as well as its cooling support.

- This is a **basis for cost savings** because subscribers can **use the service by paying very nominal**. The **provider** on the other hand **can offer the service at nominal fee to subscribers** because of **their volume of business**.

3. Reduced System Management Responsibility

- **Cloud computing model shifts majority of the infrastructure and other system management tasks towards cloud vendors**. Dedicated teams at the vendor's end takes care of all of these activities.
- Thus, **the users can enjoy a sense of relief and can concentrate only on their area (layer) of computing interest** without bothering about the management of the underlying computing layers.

4. Use-basis Payment Facility

- **Cloud computing does not charge** its subscribers **when they do not use it**.
- Even **the charge is not fixed**; it depends **on the duration of usage**.
- Rather, any **use is metered and users are charged** a reasonable fee according to **their consumption**. This reduces the cost of computing.

5. Unlimited Computing Power and Storage

- In cloud computing, users can easily **access supercomputer like computing power** at reasonable cost, if necessary. Earlier **in traditional approach**, only **big corporate** could **afford high-end computing**.
- Storage is another important issue for users. Cloud provides **as much storage as required**.
- It is **virtually unlimited** which is viewed as a big benefit for users.

6. Quality of Service

- In **traditional computing**, enterprises often used to outsource **major portion of computing related jobs to some third party**. Thus, **service quality** was broadly **dependent on the expertise** of those **third parties** or the **in-house teams** managing it.
- Whereas in cloud computing, high **quality of service (QoS)** is **ensured** as it is provided by **renowned computing vendors** having well-trained staffs and expertise exclusively in the field of computing.

7. Reliability

- The ability to deliver the **quality service** and **support load balancing, backup and recovery in cases of failure** makes the reputed cloud **vendors highly reliable** which often emerges as big worry in traditional computing.
- In cloud computing, **subscribers** no more need to **plan for all of these complex tasks** as **vendors take care** of those issues and they do it better.

8. Continuous Availability

- Reputed **cloud vendors assure almost 24 × 7 service** availability. Statistics have shown that service uptime (delivered from reputed vendors) counted for a year generally doesn't go below 99.9%.
- Such guaranteed **continuous availability** of cloud service is a big enabler for any business.

9. Locational Independence/Convenience of Access

- Cloud computing is **available everywhere via Internet**. Users **can access** it through any **computing device like PCs, or portable computing devices like tablet, laptop or smart phone**.
- **Only the thing required to avail cloud computing** through those devices is the **access to Internet**, irrespective of geographic location or time zone.

10. High Resiliency

- **Resiliency** is the **ability of reducing the magnitude and/or period of disruptions** caused by undesirable circumstances.
- Cloud computing is developed based on resilient computing infrastructure, and thus cloud services **are more resilient to attacks and faults**.
- **Infrastructure resiliency is achieved through infrastructure redundancy** combined with effective mechanism to anticipate, absorb and adapt

11. Quick Deployment

- **Deployment time in cloud environment has significantly reduced** than what is was in traditional computing environment. This is possible since **resource provisioning is rapid and automatic in cloud environment**.
- In a highly competitive market, the ability of quicker deployment gains significant business advantages.

12. Automatic Software Updates

- The issue of **software upgrade incurs a lot of headache** in traditional computing environment.
- **New patches are released** every now and then and **users need to run those patches periodically**.
- In **cloud computing environment**, this **upgrade happens automatically**.
- Cloud vendors always deliver the latest available version of any software (if not asked for otherwise).
- Upgraded environment gets available to users almost immediately after it releases, and whenever user logs in next time.

13. No License Procurement

- Application **license procurement** needed **separate budgetary arrangements** in traditional computing.
- Moreover, **unnecessary applications** used to be provided **with licensed packages**.
- Cloud computing has eliminated that problem too

14. Safety against Disaster

- **Breakdown of systems** due to sudden technical failure or natural disaster is a major concern for users.
- Specially, any **damage to physical storage** devices may **cause huge commercial loss**.

- Cloud computing delivered by reputed vendors **have robust recovery systems** incorporated in their set up. Thus, **systems and data remain more protected in cloud computing** in terms of safety and security than previous ones

15. Environment Friendly

- **Cloud computing** promotes **green computing**. **Proper utilization of resources minimizes** overall electronic resource requirement, hence reduces generation of e-waste too.
- This is beneficial for environment as **e-wastes are harmful for eco-system if not being processed properly**.
- Apart from this, the **reduced resource requirement results in lesser demand**, hence production of computing resources.
- This **decrease in e-production reduces carbon emissions** and helps to **decrease the overall carbon footprint**

III CHALLENGES OF CLOUD COMPUTING

Cloud computing provides numerous benefits. But, like any new technology, this model of computing also brings some challenges with it.

1. Limited Portability between Cloud Providers
2. Inter-operability Problem
3. Data Security
4. Reduced Control over Governance
5. Multi-Regional Compliance and Legal Issues
6. Bandwidth Cost

1. Limited Portability between Cloud Providers

- Different vendors are coming up with cloud computing facility for public use which is mostly proprietary to various extents.
- Applications developed on these proprietary clouds are difficult to move to other cloud platform due to vendor lock-in.
- This problem limits portability of applications. Hence, many times it becomes a challenge to move from one cloud provider to another

2. Inter-operability Problem

- Interoperability is the ability of a system to work with other systems.
- The proprietary issue discussed above not only raises portability problem, it also restricts applications of two different clouds to interoperate with each other. This is known as the problem of interoperability.
- Applications of two different proprietary clouds do not interoperate since they follow different standards.
- Subscribers may find two different applications from two different cloud vendors suitable for their requirement.
- For example, some enterprise may like payroll management application of one cloud vendor while accounts management application of another. But it is difficult to establish link between these two applications if they are not interoperable. Efforts have been undertaken to solve this problem

3. Data Security

- In cloud computing, users or enterprises need to store data outside their network boundary protected by firewalls.
- Thus the trust boundary of enterprises expands up to the external cloud.
- Security of users' data largely depends on the cloud vendors.
- This may introduce some extent of vulnerabilities to the security of data.
- Another concern arises when a cloud computing facility accessed by multiple parties causes overlapping of trust boundaries.

4. Reduced Control over Governance

- Cloud computing is built and governed by the policies of computing vendor or service provider.
- Consumers are relieved of the tiring responsibility of managing the computing system.
- While this turns out as a major benefit, the low control over the governance or authority of computing environment sometimes raises concerns among consumers who used to enjoy full control over self- owned traditional data centers.
- The main concern is regarding how a vendor operates the cloud.
- Although low but a certain degree of operational control is given to the subscribers depending on the type of service and service level agreement plays an important role in this regard.

5. Multi-Regional Compliance and Legal Issues

- Cloud computing vendors build data centers at locations of their convenience, both geographical and economical. A vendor may even have more than one data centers dispersed over multiple geographic locations.
- Since subscribers remotely access cloud computing over the Internet, they may not be aware of the actual location of the resources they consume. More importantly, the storage location of subscriber's data may not be within the country or region of the subscriber.
- This sometimes poses serious legal concerns.
- Most regulatory frameworks recognize cloud consumer organizations responsible for the security, integrity, and storage of data even when in reality it is held by an external cloud vendor.
- In such scenario, resolving the multi-regional compliance and legal issues are soaring challenges before cloud computing.

6. Bandwidth Cost

- But it is a fact that whiles the pay-per-use model of cloud computing cuts down costs as subscribers only pay for the resources or services they use, the model brings some associated cost along with it. That is the cost of network bandwidth used to access the service.
- In the current age of Internet, cost of bandwidth is very low at moderate speed of access.
- But more bandwidth can provide higher speed which is essential for high quality service.
- While low cost bandwidth may often fulfill requirements of general applications, data intensive applications (those deal with critical and huge volume of data sets) demand higher bandwidth which may add a little more in the total cost of computing.

IV CLOUD COMPUTING SERVICES

Cloud computing environment is built of heterogeneous computing systems spread over different geographic locations. Together they all act as a single system. Applications running on those disparate systems communicate with each other through the web services.

A web service is the way of establishing communication between two software systems over the internetwork. Web services use standardized way of data exchange since different software systems might be built using different programming languages and run on different platforms.

Thus, this standardization is very important so that communication remains independent of programming languages or platforms



FIG 3.3: The cloud from the point of view of users

Web Service

Web service describes the method of establishing communication between two web-based applications. World Wide Web Consortium (W3C) defines web services as “a software system designed to support interoperable machine-to-machine interaction over a network”.

Web services are generally categorized into two different classes. Based on how the service is being implemented, a web service can either be

1. Simple Access Object Protocol (SOAP) based,
2. Representational State Transfer (REST) compliant

1. SOAP-Based Web Services:

SOAP-based web services use XML format for communicating messages among web applications as XML is an open format and recognized by all kind of applications.

In this approach, HTTP or hyper-text transfer protocol is used for passing messages.

The SOAP is originally developed by Microsoft as older Remote Procedure Call (RPC)- based message passing technologies like DCOM (Distributed Component Object Model) or CORBA (Common Object Request Broker Architecture) did not work well with Internet.

This was primarily because those technologies relied on binary messaging. On the other hand, the XML format of messaging performs well over Internet.

The SOAP was accepted as standard when Microsoft submitted it to the Internet Engineering Task Force (IETF).

The rules of SOAP communications are described in Web Services Description Language (WSDL) format. The rules are stored in files with .wsdl extension.

SOAP is often considered as complex since creation of the XML structure is mandatory for passing message

2. REST: Compliant Web Services

REST represents a simpler way of communicating messages.

REST provides a light weight alternative.

Instead of using XML to make a request, REST relies on global identifier to locate the resources. Thus, a separate resource discovery mechanism is not needed.

This 'representation' of paths of applications provides the additional power in REST.

The global identifier assigned to each resource makes the access method of the resources uniform. Any resource at a known address can be accessed by any application agents.

REST allows many standard formats like XML, JavaScript Object Notation (JSON) or plain text as well as any other agreed upon formats for data exchange.

REST is an architecture style for designing networked applications. Here, simple HTTP is used to make calls between machines identified by their URLs (Uniform Resource Locators) which is simpler than other mechanisms like CORBA, DCOM or SOAP

SOAP versus REST

Both SOAP and REST can be used for cloud computing development. The choice depends on the design and implementation requirements.

Table 3.2 Differences between SOAP and REST

<i>SOAP</i>	<i>REST</i>
It is a message communication protocol that is used to build network application.	It is an architectural style for network application.
It uses XML for formatting message.	It uses simpler messaging formats like JSON. It also supports XML.
It requires the XML parsing.	Here, the XML parsing can be avoided by using JSON.
SOAP defines its own additional security over HTTPS.	It inherits security measures from the underlying transmission protocol as HTTPS.
It is a bit heavy.	It is lighter than SOAP and response time is better.
It is complex than REST.	It is simple to use and easy to understand.
It is recognized as a standard protocol.	It is not an standard, rather an architectural style.
SOAP uses services interfaces to expose the business logic.	REST uses Uniform Resource Identifier (URI) to expose business logic.
SOAP oriented development is faster since it is supported by many tools.	Development time is longer due to limited tool support.
It shows lesser flexibility in controlling resources.	It shows more flexibility in controlling resources.
It requires less knowledge in programming.	It requires rather greater knowledge in programming in comparison with SOAP system.

ROLE OF API:

API (Application Program Interface) is a set of defined functions or methods which is used to compile the application.

It defines the contract of communication or standard interface provided by software components for others (other software components) in order to interact with them.

APIs play important role in cloud computing. When some cloud services are released, corresponding APIs (referred as cloud API) are also released as they are critical for the usefulness and operational success of those services. Cloud services generally provide well- defined APIs for its consumers so that anyone can access and use the capabilities offered to develop application or service.

Request for data or computation can be made to cloud services through cloud APIs. Cloud APIs expose their features via REST or SOAP.

For instances, the cloud APIs can be classified as

- IaaS (Infrastructure as a Service) API for resource configuration or workload management,
- PaaS (Platform as a Service) API to integrate with database service etc.
- SaaS (Software as a Service) API to integrate with application services like ERP or CRM.

Both vendor specific and cross platform APIs are available, but cross platform APIs are still not widespread in cloud computing arena and are available for specific functional areas only.

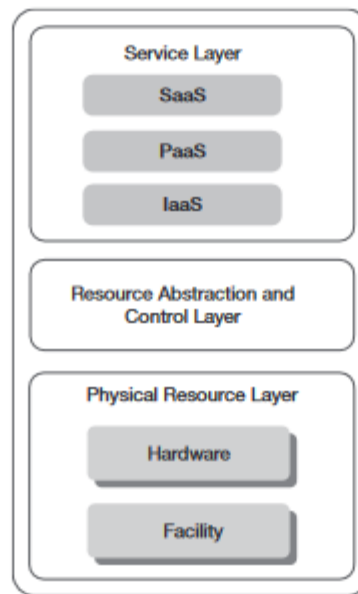


FIG 5.1: Service Layer as part of Cloud Service Orchestration⁶

Clubbed together these three service models are commonly referred as SPI (Service-Platform-Infrastructure) model. Cloud service providers arrange these services for the cloud consumers. The NIST reference architecture represents these services under ‘service orchestration’ component of the providers (Figure 5.1)

Service layer is the topmost layer of cloud service orchestration over the resource abstraction and control layer. The service layer includes three major cloud services as SaaS, PaaS and IaaS. The PaaS layer resides over IaaS and SaaS layer resides over PaaS.

Infrastructure-as-a-Service

- Cloud computing allows access to computing resources in a virtualized environment popularly referred as ‘the Cloud’. Infrastructure-as-a-Service delivers virtualized-hardware (not physical, but simulated software) resources to consumers known as virtual resources or virtual components.
- It provides the facility of remotely using virtual processor, memory, storage and network resources to the consumers.
- These virtual resources can be used just like physical (hardware) resources to build any computing setup (like virtual machine or virtual network).
- For this reason, IaaS is also referred as Hardware-as-a-Service (HaaS).
- IaaS is the bottommost layer of cloud computing service model. The IaaS component stack is shown in the below Figure .
- It is a computing solution where the complexities and expenses for managing the underlying hardware are outsourced to some cloud service providers.
- In cloud computing IaaS model, provider arranges and manages hardware resources for users but users cannot access those hardware resources directly.
- The hardware resources are represented as simulated software components and are delivered to consumers via Internet (or network) using web services.

- Consumers can use those simulated components just like real hardware devices and can build computing system with necessary processor, memory, storage and network facilities.

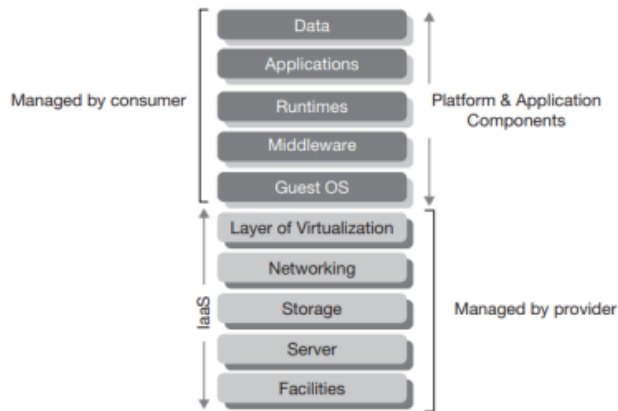


FIG 5.2: IaaS component stack

Platform-as-a-Service

- In computing, platform means the underlying system on which software applications can be installed
- A computing platform comprises hardware resources, operating system, middleware (if required) and runtime libraries. Application programs are also installed over this platform.
- Application development and deployment in traditional computing require the users' participation in managing hardware, operating system, middleware, web servers and other components.
- For instance, users must install appropriate framework (like J2EE, .NET) before working in any application platform.
- PaaS facility, on the other hand, relieves users from all these tensions and delivers ready-made platform to consumers via internetwork/Internet.
- PaaS component stack, in addition, provides application (development and deployment) platform over IaaS component stack.
- A PaaS provider not only delivers fully-managed application development and deployment environment but also takes care of the lower level (infrastructure level) resource management and provisioning

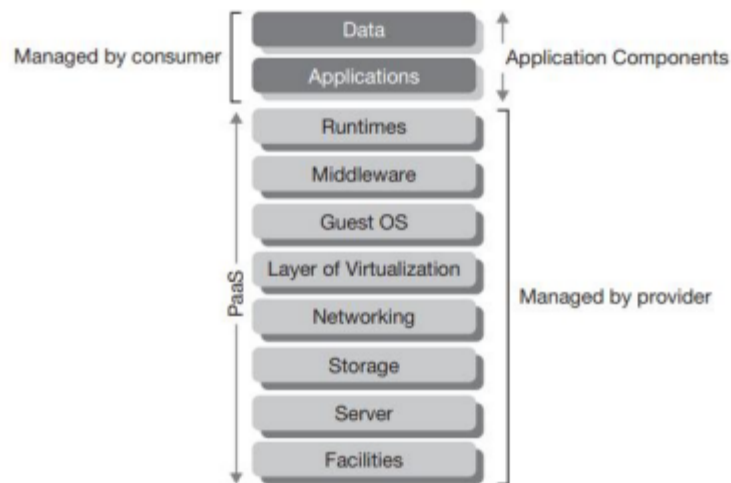


FIG 5.3: PaaS component stack

- Thus, PaaS is created by adding additional layers of software over IaaS. With the use of PaaS, collaborative application development becomes easier where multiple users can work from different geographical locations.
- PaaS also reduces the total cost of ownership (TCO) as computing platform becomes available on rent basis.
- There are many PaaS offerings available in market. Google App Engine, Microsoft Azure Platform, GoGrid Cloud Center, Force.com are very popular among them. Open-source PaaS offerings are also available in the market. Cloud foundry is one such which is developed by VMware

Software-as-a-Service

- Software-as-a-Service (SaaS) is a way of delivering application as a service over the network/Internet that users can directly consume without the tension of installing or configuring an application.
- In traditional computing, consumers had to pay not only the software licensing fee but also spend a large portion of their budget in setting up the infrastructure and platform over which the application would run. SaaS eliminates this problem and promises easier as well as a cheaper way of using application.
- SaaS is hosted by SaaS vendors and delivered to the consumers over network/Internet. Unlike traditional packaged applications that users install on their own computing setup, SaaS vendors run it in their data centers.
- SaaS applications are sometimes referred as web-based software, or hosted software.
- In SaaS model, everything is managed by vendor including application upgrade or updates even the data and application acts upon are also managed by SaaS .

Example(any one or two):

1. E-mail facility is one common example of SaaS application that is used by everyone.

2. The CRM (customer relationship management) package of Salesforce. com gained popularity among enterprises since early 2000s.
3. SAP (Systems, Application and Products) as the solution provider of Enterprise Resource Planning (ERP) entered into the SaaS CRM and ERP markets with its 'Business ByDesign' solution.
4. Oracle launched its CRM SaaS 'On Demand'. There are also many popular SaaS offerings for general users in the market today like GoogleApps, Microsoft Office 365 and else.

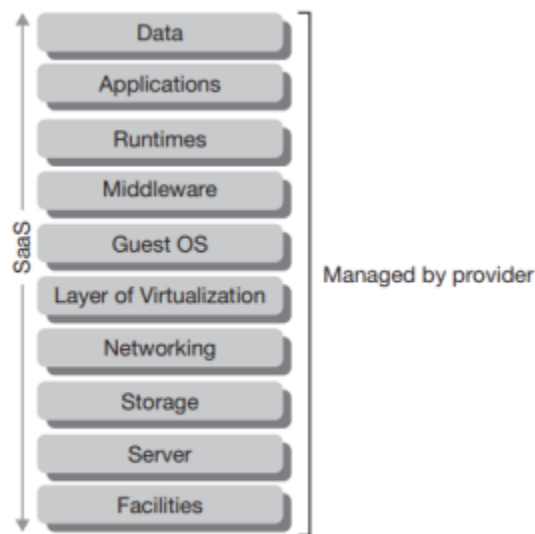


FIG 5.5: SaaS component stack

SERVICE ABSTRACTION

- The word 'abstraction' is derived from Latin words 'abs' meaning 'away from' and 'trahere' meaning 'to draw'. Abstraction takes away or hides the characteristics in order to reduce something into an essential utility.
- In computing, layer of abstraction is a way of hiding the implementation details of a particular set of functionalities. In cloud computing model, the level of service abstraction raises while moving from IaaS towards SaaS.
- At the IaaS level, consumers have the freedom of building computing infrastructure (essentially in virtual mode) from a set of available options.
- They can configure machines, setup network and select storages. All of the devices are available in virtual mode and consumers should have the knowledge of building the required infrastructure from the scratch. In PaaS level, the abstraction intensifies. The underlying virtual infrastructure of IaaS level remains hidden from the consumers.
- They can simply work with chosen platform environment without knowing how the underlying system has been configured.
- At this level, consumers have full control over the PaaS environment and they can configure and manage the utilities like web servers, database servers, and different application development

environments. This abstraction intensifies further at SaaS level where consumers remain unaware even about the technology of an application. They simply use applications without knowing anything about its implementation

THE SPI MODEL

The three service models SaaS-PaaS-IaaS together is referred as the SPI model of cloud computing.

In this layered architecture of cloud service model, by moving upward, each service layer is empowered by the capabilities of the service layer(s) beneath it.

Thus, PaaS is empowered by IaaS layer and SaaS is empowered with the potentials of underlying IaaS and PaaS layers. Figure below shows the layered cloud SPI model with the mention of popular commercial services.

One point to note here is that there is no functional relation between cloud deployment models and delivery models.

Any delivery model can exist in any deployment scenario and thus any delivery/deployment pairing is possible. Although, SaaS offerings are mostly public services and hence the SaaS/public combination is more common than the other combinations.

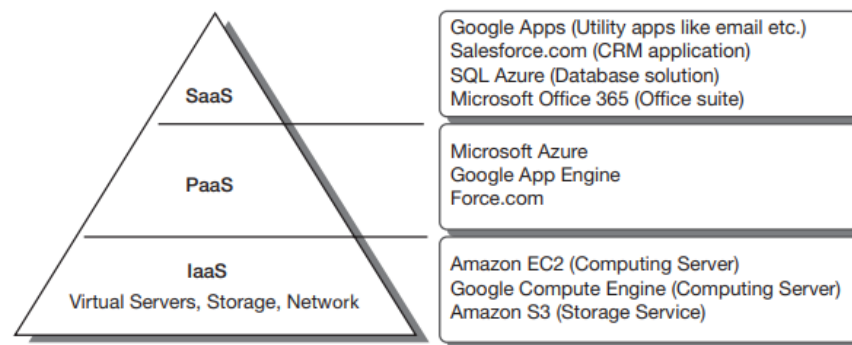


FIG 5.8: The layered cloud service model

UBIQUITOUS CLOUD

The term 'ubiquitous' means as being everywhere.

The idea of ubiquitous computing talks about making computing facility available everywhere and for all the time.

The ubiquitous computing paradigm is also known as pervasive computing.

In comparison with desktop computing, interaction with computing environment in ubiquitous computing can happen using any devices having some computing capability of any form.

Cloud computing further strengthens the idea of ubiquitous computing. Ubiquitous cloud refers to the use of computing resources spread over geographic locations from any place and any time.

CONFUSION BETWEEN CLOUD AND INTERNET

The actual fact is that earlier users used to access websites or web portals consisting of static and dynamic pages via Internet but now they access cloud computing too.

Cloud computing has its own characteristics. It follows utility service model and it is measurable where users can be billed as per use. More importantly, it can deliver both software and hardware to users over internetwork or Internet in special form called 'service'.

Cloud computing does not mean simple static or dynamic web content; it is much more than that.

V RESOURCE VIRTUALIZATION

WHAT IS VIRTUALIZATION

- Virtualization refers to the representation of physical computing resources in simulated form having made through the software. This special layer of software (installed over active physical machines) is referred as layer of virtualization.
- This layer transforms the physical computing resources into virtual form which users use to satisfy their computing needs.
- Figure below represents the basic concept of virtualization in simplified form which has been elaborated in the subsequent sections.

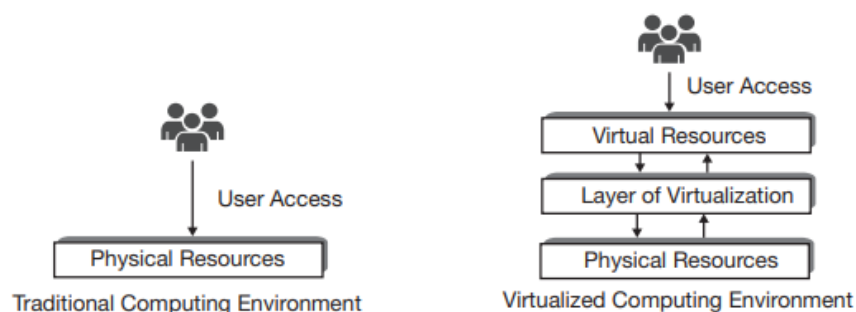


FIG 7.1: Users' interaction with computer in traditional and virtualized computing environment

- In simple sense, the virtualization is the logical separation of physical resources from direct access of users to fulfill their service needs.
- Although, at the end, actually the physical resources are responsible to provide those services. The idea of virtualizing a computer system's resources (including processor, memory, storage etc.) has been well-established since many decades.
- Virtualization provides a level of logical abstraction that liberates user-installed software (starting from operating system and other systems as well as application software) from being tied to a specific set of hardware.

- Rather, the users install everything over the logical operating environment (rather than physical ones) having created through virtualization.

VIRTUALIZING PHYSICAL COMPUTING RESOURCES

Any kind of computing resources can be virtualized. Apart from basic computing devices like processor, primary memory, other resources like storage, network devices (like switch, router etc.), the communication links and peripheral devices (like keyboard, mouse, printer etc.) can also be virtualized.

But, it should be noted that in case of core computing resources a virtualized component can only be operational when a physical resource empowers it from the back end.

For example, a virtual processor can only work when there is a physical processor linked with it. Figure below represents a virtualized computing environment comprising of processor, memory and storage disk.

The layers of virtualization transforms these physical computing devices into virtual form and presents them before user.

The important thing here to be noted is that the simulated devices produced through virtualization may or may not resemble the actual physical components (in quality, architecture or in quantity).

For instance, in the given figure below, users get access to three processors while there is one physical processor in reality. Or, 32-bit processor can be produced (in virtual form) from 64-bit actual physical processor.

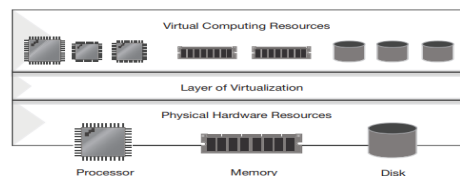


FIG 7.2: Transformation of physical resources into virtual resources through virtualization

- The software for virtualization consists of a set of control programs. It offers all of the physical computing resources in custom made simulated (virtual) form which users can utilize to build virtual computing setup or virtual computers or virtual machines (VM).
- Users can install operating system over virtual computer just like they do it over physical computer. Operating system installed over virtual computing environment is known as guest operating system.
- When virtualization technique is in place, the guest OS executes as if it were running directly on the physical machine.

MACHINE OR SERVER LEVEL VIRTUALIZATION

- Machine virtualization (also called server virtualization) is the concept of creating virtual machine (or virtual computer) on actual physical machine.

- The parent system on which the virtual machines run is called the host system, and the virtual machines are themselves referred as guest systems.
- In conventional computing system, there has always been a one-to-one relationship between physical computer and operating system. At a time, a single OS can run over them.
- Hardware virtualization eliminates this limitation of having a one-to-one relationship between physical hardware and operating system.
- It facilitates the running of multiple computer systems having their own operating systems on single physical machine.
- As shown in Figure, the OS of the guest systems running over same physical machine need not to be similar. All these virtual machines running over a single host system, remain independent of each other. Operating systems are installed into those virtual machines.
- These guest systems can access the hardware of the host system and can run applications within its own operating environment.

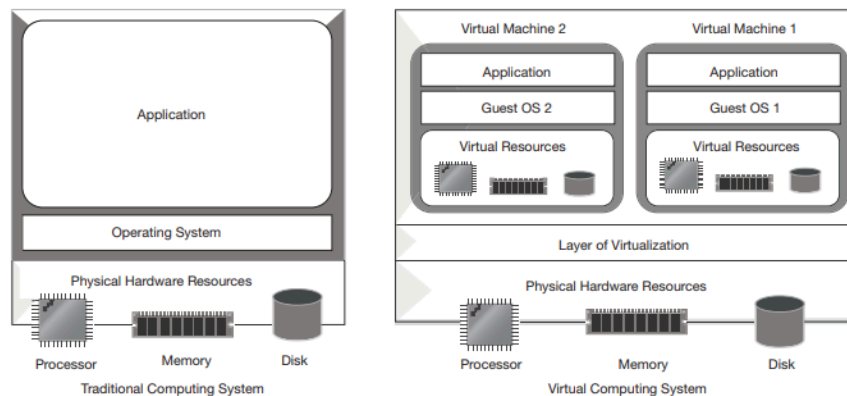


FIG 7.3: Conventional computing system versus Virtualized computing system

The Layer of Virtualization

- From above Figure it can be seen that virtual machines are created over the virtualization layers. This virtualization layer is actually a set of control programs that creates the environment for the virtual machines to run on.
- This layer provides the access to the system resources to the virtual machines. It also controls and monitors the execution of the virtual machines over it. This software layer is referred as the Hypervisor or Virtual Machine Monitor (VMM).
- The hypervisor abstracts the underlying software and/or hardware environments and represents virtual system resources to its users.
- This layer also facilitates the existence of multiple VMs those are not bound to share same (underlying) OS kernel.

Machine Virtualization Techniques

- There are two different techniques of server or machine virtualization as hosted approach and the bare metal approach.
- The techniques differ depending on the type of hypervisor used.

- Although the techniques are different but they have the same end or ultimate goal by creating a platform where multiple virtual machines can share same system resources. Each technique is simply a different way of achieving this goal.

1.Hosted Approach

- In this approach, an operating system is first installed on the physical machine to activate it.
- This OS installed over the host machine is referred as host operating system. The hypervisor is then installed over this host OS. This type of hypervisor is referred to as Type 2 hypervisor or Hosted hypervisor.
- Figure below represents the hosted machine virtualization technique. So, here the host OS works as the first layer of software over the physical resources.
- Hypervisor is the second layer of software and guest operating systems run as the third layer of software. Products like VMWare Workstation and Microsoft Virtual PC are the most common examples of type 2 hypervisors.

Benefits: In this approach, the host OS supplies the hardware drivers for the underlying physical resources. This eases the installation and configuration of the hypervisor. It makes the type-2 hypervisors compatible for a wide variety of hardware platform.

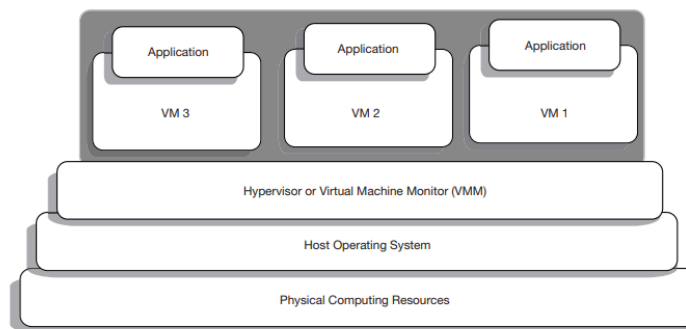


FIG 7.4: A model of hosted machine virtualization approach

Drawbacks: A hosted hypervisor does not have direct access to the hardware resources and hence, all of the requests from virtual machines must go through the host OS. This may degrade the performance of the virtual machines

2.Bare Metal Approach: Removal of the Host OS

- In this approach of machine virtualization, the hypervisor is directly installed over the physical machine. Since, the hypervisor is the first layer over hardware resources, hence, the technique is referred as bare metal approach.
- Here, the VMM or the hypervisor communicates directly with system hardware. In this approach, the hypervisor acts as low-level virtual machine monitor and also called as Type 1 hypervisor or Native Hypervisor. VMware's ESX and ESXi Servers, Microsoft's Hyper-V, solution Xen are some of the examples of bare-metal hypervisors.

Benefits: Since the bare metal hypervisor can directly access the hardware resources in most of the cases it provides better performance in comparison to the hosted hypervisor. For bigger application like enterprise data centers, bare-metal virtualization is more suitable because usually it provides advanced features for resource and security management. Administrators get more control over the host environment.

Drawbacks: As any hypervisor usually have limited set of device drivers built into it, so the bare metal hypervisors have limited hardware support and cannot run on a wide variety of hardware platform

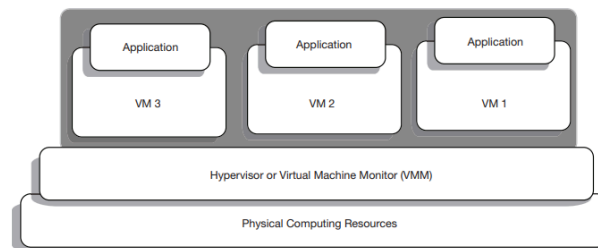


FIG 7.5: A model for the bare metal approach of machine virtualization

Type 1 or bare metal hypervisor does not use any host operating system.

EXPLORING HYPERVISOR OR VIRTUAL MACHINE MONITOR

- The hypervisor or virtual machine monitor (VMM) presents a virtual operating platform before the guest systems. It also monitors and manages the execution of guest systems and the virtual machines.
- All of the virtual machines run as self-sufficient computers isolated from others, even though they are served by the same set of physical resources.
- Alternately, it can be said that a hypervisor or VMM facilitates and monitors the execution of virtual machines and allows the sharing of the underlying physical resources among them. Following section focuses on different hypervisor-based virtualization approaches.

Hypervisor-Based Virtualization Approaches

Hypervisor-based virtualization techniques can be divided into three categories as full virtualization, para-virtualization and hardware-assisted virtualization. The techniques being used in these three approaches are entirely different from one another.

1. Full Virtualization

- In full virtualization (also called as native virtualization), the hypervisor fully simulates or emulates the underlying hardware. Virtual machines run over these virtual set of hardware.
- The guest operating systems assume that they are running on actual physical resources and thus remain unaware that they have been virtualized.

- This enables the unmodified versions of available operating systems (like Windows, Linux and else) to run as guest OS over hypervisor. In this model, it is the responsibility of the hypervisor to handle all OS-to-hardware (i.e. guest OS to physical hardware) requests during running of guest machines.
- The guest OS remains completely isolated from physical resource layers by the hypervisor. This provides flexibility as almost all of the available operating systems can work as guest OS. VMware's virtualization product VMWare ESXi Server and Microsoft Virtual Server are few examples of full virtualization solution.

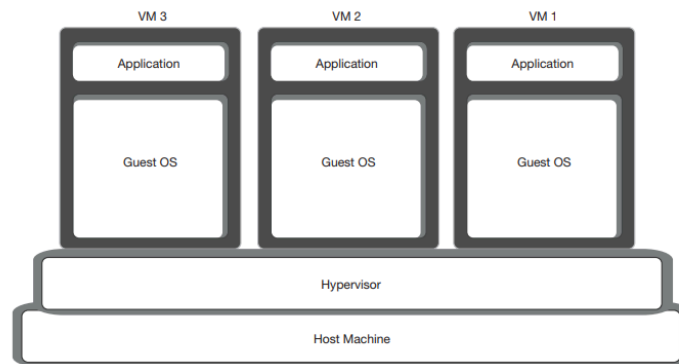


FIG 7.6: A model of full virtualization

2 .Para-Virtualization or OS-Assisted Virtualization

- 'Para' is an English affix of Greek origin that means 'beside' or 'alongside'. In full virtualization, the operating systems which run in VMs as guest OSs need not to have any prior knowledge that they will run over virtualized platform.
- They do not need any special modification (or, functionality incorporation) for running over the hypervisors and are installed in their original form.
- The whole virtualization activities are managed by the hypervisor like translating instructions and establishing communication between different guest OSs and the underlying hardware platform.
- In para-virtualization, a portion of the virtualization management task is transferred (from the hypervisor) towards the guest operating systems.
- Normal versions of available operating systems are not capable of doing this. They need special modification for this capability inclusion. This modification is called porting.
- Each guest OS is explicitly ported for the para-application program interface (API). A model of para-virtualization has been shown in Figure .
- Thus, in para-virtualization, each guest OS needs to have prior knowledge that it will run over the virtualized platform.
- Moreover, it also has to know on which particular hypervisor they will have to run. Depending on the hypervisor, the guest OS is modified as required to participate in the virtualization management task.

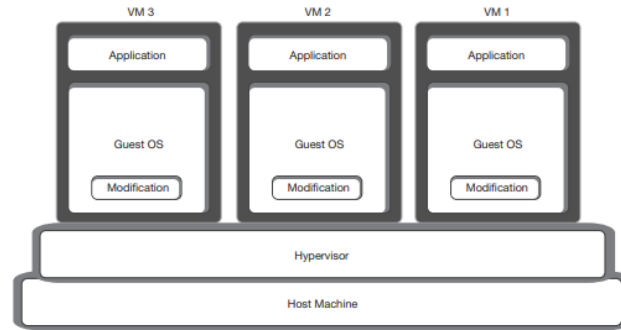


FIG 7.7: A model of para-virtualization

- The unmodified versions of available operating systems (like Windows, Linux) cannot be used in para-virtualization. Since it involves modifications of the OS the para-virtualization is sometimes referred to as OS-Assisted Virtualization too.
- This technique relieves the hypervisor from handling the entire virtualization tasks to some extent. Best known example of paravirtualization hypervisor is the open-source Xen project which uses a customized Linux kernel.

3. Hardware-Assisted Virtualization

- Inspired by software-enabled virtualization, hardware vendors later started manufacturing devices tailored to support virtualization. Intel and AMD started this by including new virtualization features in their processors.
- The AMD-Virtualization (AMD-V) and Intel Virtualization Technology (Intel-VT) allows some privileged CPU calls from the guest OS to be directly handled by the CPU. These calls do not require to be translated by the hypervisors.
- Many bare-metal hypervisors make use of this technology. Hypervisors like Xen, Microsoft's Hyper-V or VMWare ESXi Server can take the advantages of the hardware-assisted virtualization.

Table 7.3 Comparison among Full Virtualization, Para-Virtualization and Hardware-Assisted Virtualization

<i>Full Virtualization</i>	<i>Para-Virtualization or OS-Assisted Virtualization</i>	<i>Hardware-Assisted Virtualization</i>
Guest OS has no role in virtualization.	Guest OS plays role in virtualization.	Guest OS has no role in virtualization.
Guest OS remains unaware about the virtualization.	Guest OS has to be aware about the virtualization.	Guest OS remains unaware about the virtualization.
Normal version of available OS can be used as guest OS.	Modified version of available OS is required.	Normal version of available OS can be used as guest OS.
It provides good options for guest OS.	It provides lesser options for guest OS.	It provides good options for guest OS.
Guest OS is not hypervisor-specific.	Guest OS is tailored to be hypervisor-specific.	Guest OS is not hypervisor-specific.
Here it requires no special feature in the host CPU.	Here it requires no special feature in the host CPU.	Here it requires explicit features in the host CPU.
Hardware does not play role in virtualization.	Hardware does not play role in virtualization.	Hardware plays role in virtualization.
Hypervisor takes care of all of the virtualization tasks.	Guest OS along with hypervisor take care of the virtualization tasks.	Specialized hardware device along with hypervisor take care of virtualization tasks.
Virtualization overhead of hypervisor is more.	Virtualization overhead of hypervisor is less.	Virtualization overhead of hypervisor is less.
Virtualization performance is little slow.	Virtualization performance is better.	Virtualization performance is better.
It provides high level of security as all of the virtualization controls remain with the hypervisor.	Here the security is compromised as guest OS has some control in virtualization.	Here the security is compromised as calls from guest OS can directly access the hardware.

Threat to Hypervisor:(vulnerabilities of the Hypervisor)

The hypervisor runs virtual machines and largely protects the underlying physical resources from damage. System designers should be aware of those hitches as any threat to the hypervisor may turn the entire system vulnerable.

Following section focuses the common vulnerabilities of the hypervisor

1. Rogue Hypervisor:

2. VM Escape

Rogue Hypervisor:

- The system can be completely hacked by Hackers if they create rogue hypervisor (hypervisor made with wrong intention) and can replace an original hypervisor with the same in order to attack a system without any knowledge of the system.
- Since hypervisor is the prime medium of communication between host machine and virtual machines, the rogue hypervisor can easily be used to cause damage to a system.

- This is similar to a root-kit infection of a physical computer. Rootkit is a malicious piece of software inserted by an intruder into a computer to serve some illegal purpose without being detected.
- The key for rootkit is the access to the system root (in context with Unix OS).
- Rootkits themselves are not harmful or do not cause any harm to the system directly.
- They are used to hide entry of malware or worms into the system by creating some sort of hidden channels.
- Virtualization-based rootkits in the same way can use a rogue hypervisor to create a covert channel to dump unauthorized code into a system

VM Escape:

122Cloud Computing

7.6.2.1 Rogue Hypervisor

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Virtualization-based rootkits in the same way can use a rogue hypervisor to create a covert channel to dump unauthorized code into a system.

Rogue hypervisor may create covert channel for malware to attack the system.

7.6.2.2 VM Escape

One advantage of running applications in virtual machine is that :

- The applications as well as the users cannot get direct access to the hypervisor or the host operation system.
- Hypervisor creates isolated environments for different virtual machines running on it. So, applications running on one virtual machine cannot affect other VMs.

But, an improperly configured or manipulated virtual machine may allow codes to completely bypass this security and enable attackers to gain access to the hypervisor.

Such bypass is known as **virtual machine escape** where the exploiter can directly access the hypervisor or host operating system and can even perform administrative tasks like VM creation, deletion, manipulation etc.

VM escape may result in a complete failure to set up the security mechanism of the VM's physical host.

The threats as mentioned above to hypervisor can be eliminated with

- Attentive system administration.
- A properly managed hypervisor can even protect virtualized system against threats cropping up from any weakness in operating system of the VMs.

So, while implementing virtualization, proper measures must have to be taken against the common threats to the hypervisor.

OPERATING SYSTEM LEVEL VIRTUALIZATION: REMOVAL OF THE HYPERVISOR

Operating system level virtualization also called as system level virtualization

The host-guest paradigm does not work.

The virtual machine terminology is also not used here.

No hypervisor is used

The virtual servers are enabled by the kernel of the operating system of physical machine.

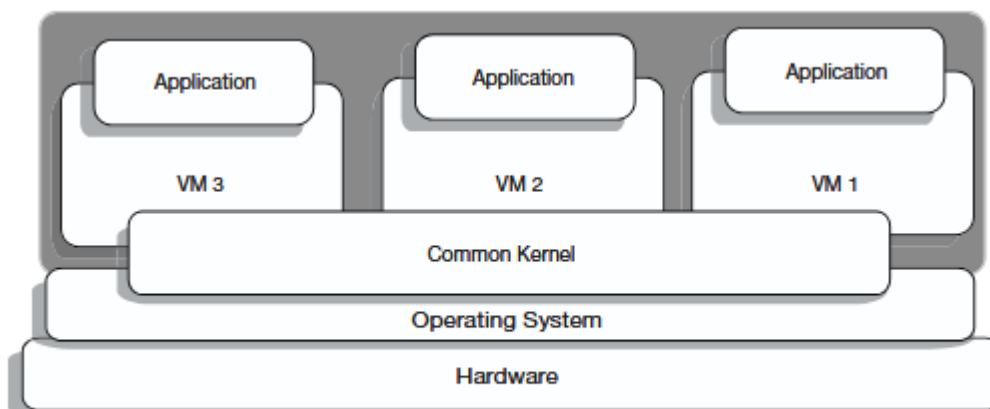


FIG 7.8. A model of operating system-level virtualization approach

In this approach:

The kernel of the operating system installed over physical system is shared among all of the virtual servers running over it.

Since all of the virtual servers share a single kernel, means that all of them will have same OS as the parent system.

The goal of this approach is to create multiple logically-distinct user-space instances (virtual servers) over a single instance of an OS kernel.

This approach is also known as Operating System Virtualization or Shared Kernel Approach.

Example FreeBSD's jail, Linux VServer, OpenVZ

All of them can run logically-distinct user-spaces on top of a single kernel.

Advantages: The advantages of OS level virtualization is that it is lighter in weight since all of the virtual servers share a single instance of an OS kernel.

This enables a single physical system to support many others virtual servers than the number of complete virtual machines it could support.

Limitations: All virtual machines have to use the same operating system (due to sharing of OS kernel).

Operating System level virtualization	Machine virtualization
Here virtual server share single kernel and can run only on a similar os	In this one can different os in different virtual system running on same physical resource

MAJOR SERVER VIRTUALIZATION PRODUCTS AND VENDORS

Companies like VMware, Citrix and Microsoft are some of the leading vendors of server virtualization products.

- **VMware vSphere:**
- VMware is a pioneer in the virtualization market especially in server and desktop virtualization.
- This U.S.-based company was founded in 1998 and has released many products for computer infrastructure virtualization.
- vSphere is VMware's enterprise-level server virtualization product.
- It is one of the industry's leading and reliable virtualization platforms and is used in data center virtualization for cloud computing.
- The vSphere was released as an enhanced suite of its earlier product VMware Infrastructure (also known as VI).
- vSphere uses ESXi hypervisor which is also a product of VMware.
- This hypervisor is Type 1 or bare-metal hypervisor which gets installed directly on the physical server.

- vSphere software suite also provides a centralized management application called as vCenter Server through which virtual machines and ESXi hosts can be managed centrally.

Citrix XenServer:

- XenServer is one of the leading open-source server virtualization solutions offered by U.S. software company Citrix Systems.
- Xen hypervisor which has pioneered the para-virtualization concept, is the basis of XenServer.
- The Xen originated as an open-source research project at the University of Cambridge and its first release was made in 2003.
- The Xen hypervisor has been used in many server virtualization solutions, both commercial and open-source.
- Amazon EC2, Oracle VM and XenServer are notable among them.
- The developer group of Xen is XenSource Inc. which later was acquired by Citrix Systems in 2007.
- Most of the features of Citrix XenServer are available free of cost.

Microsoft Hyper-V Server:

- Hyper-V Server is the server virtualization solution from Microsoft.
- First launched in 2008,
- It was formerly known as Windows Server.
- Now Hyper-V Server comes as a dedicated stand-alone product and also regarded as a feature of Windows Server.

Oracle VM VirtualBox:

- Oracle VM VirtualBox (formerly known as Sun VirtualBox) is the virtualization software package from Oracle Corporation.
- VirtualBox was purchased by Sun Microsystems from its developer innotek GmbH, in 2008 and now owned by Oracle.
- This Xen hypervisor based open-source product runs on a wide variety of host operating systems and supports a large number of guest operating systems too.

KVM:

- The kernel-based virtual machine (KVM) is a hypervisor built into the Linux kernel.
- This open-source solution was developed by Red Hat Corporation to provide virtualization services on the Linux-operating system platforms.
- It has been part of Linux kernel since version 2.6.20 and currently being supported by several distributions.
- A wide variety of guest operating systems work with KVM including several versions of Windows, Linux and UNIX.

HIGH-LEVEL LANGUAGE VIRTUAL MACHINE

The high-level language virtual machine is a concept that goes against the idea of conventional computing environment where a compiled application is firmly tied to a particular OS and ISA (instruction set architecture).

Thus, in conventional computing environment, the porting of application to different computing platform requires recompilation of the application code for the targeted platform.

porting of application needs porting of the underlying compiler for the target platform first which is a tedious technical task.

The high-level language VM eases the porting of compiler by rendering HLLs to intermediate representation targeted towards abstract machines.

The abstract machine then translates the intermediate code to physical machine's instruction set. This concept of abstract machine (as well as intermediate code representation) makes the porting of compilers less complex as only the back-end part of the compiler needs to be ported since the intermediate code representation is same for compilers of a HLL on any platform.

The intermediate representation of a HLL program having compiled over one physical computing architecture can be ported on abstract machine running over other architecture.

The abstract machine then translates the intermediate code to the instruction set of underlying architecture.

Such abstract machine is referred as high-level language VM or HLL VM. Java Virtual Machine (JVM) and Microsoft's Common Language Runtime (CLR) are two examples of high-level language VMs.

High-level language VM is also known as **application VM or process VM**.

Application or process virtualization can be considered as the smaller version of machine virtualization. Instead of virtual machines, this technology decouples application software from the underlying platform on which it is executed.

The best known example of this type of virtualization is Java Virtual Machine (JVM).

JVM makes the execution of java applications machine-independent as the applications do not interact with the underlying OS and hardware platform directly. Rather, the JVM (which are usually machine-dependent) creates a uniform virtual machine environment (over different operating system and hardware architecture) and makes java applications portable.

Microsoft has even adopted a similar approach in the Common Language Runtime (CLR) used by .NET applications. The concept of process virtualization was actually pioneered by the UCSD P-System (a portable operating system) during late 1970s which was developed by a team at the University of California at San Diego, United States.

Application virtual machine places a sort of software wrapper around the application running over it and this wrapper restricts the application from directly interacting with system.

This is referred to as **sandboxing**. The sandbox technology is often used to test unverified programs which may contain malicious code in order to prevent them from damaging the underlying hardware devices.

EMULATION:

Emulation in computing is done by making one system imitating another.

This means a system having some architecture is made enable to support instruction set of some other machine architecture.

For example, let a piece of software has been made for architecture 'A' and is not supported by architecture 'B'. Through emulation, it is possible to imitate the working of system 'A' (i.e. architecture 'A') on system 'B' (i.e. architecture 'B') and then the piece of software to run on system B. Emulators can be software or hardware both.

Emulation software converts binary data written for execution on one machine to an equivalent binary form suitable to execute on another machine. This is done by translating the binary instructions.

There are two ways for implementation of emulations like **interpretation and binary translation**.

In binary translation (also known as recompilation), a total conversion of the binary data (made for the emulated platform) is done. The conversion recompiles the whole instruction into another binary form suitable to run on the actual or targeted platform.

There are two types of binary translation like **static recompilation and dynamic recompilation**.

In interpretation, each instruction is interpreted by the emulator every time it is being encountered. This method is easier to implement but slower than binary translation process.

Simple Virtualization	Emulation based virtualization
Multiple VM with different Guest Operating System torun on single host Machine	Multiple VM with different Guest Operating System torun on single host Machine
The application running on VM is compiled for native instruction set of host machine	The application running on VM is compiled for some other architecture and instruction set
Instruction set used by virtual system and actual H/w is same	VM simulates the complete H/w and S/w
The VM simply passes the service request to actual physical System	It can run on one H/w and create environment to support some other architecture
Transaltion of instruction set is not required	This allows an OS made for one computer architecture to run on architecture supported by emulator
	Instruction set of virtual system is not same as actual physical system

The performance of VM is much faster	Hence service request received by VM can only be passed to physical system after translation of instruction set
	Emulation is slow compared to simple virtualization
	Microsoft virtual PC, QEMU (quick emulator)

Some other types of virtualization:

cloud-based service development requires each and every computing infrastructure in virtualized mode.

virtualized network infrastructure, storage systems etc. are as important as the server virtualization.

Network Virtualization

Storage Virtualization

Desktop Virtualization

Network Virtualization:

Network virtualization is the process of combining network resources and network functionality into a single, software-based administrative entity called as a virtual network.

There are two common forms of network virtualization which have been discussed below.

Virtual device-based virtual network: Here, virtualized devices form the network. All virtual networking devices (including virtual computers, virtual switches, virtual routers etc.) communicate using actual (non-virtual) network protocols such as Ethernet as well as virtualization protocols such as the VLAN. This is actual network virtualization where the network is formed with all virtual components.

Protocol based virtual network: Rather than virtualizing devices, it creates virtual area network. Virtual LAN (VLAN) and virtual private network (VPN) are examples of such virtualizations.

These are logical local area networks (logical LANs) where the underlying physical LAN's structure is something else. Here, several physical LANs which are actually part of public network (such as the Internet) can function as a single logical LAN. This enables network devices (such as computers and switches) to send and receive data across shared or public networks as if they are part of a private network. The devices can communicate using LAN protocols which make faster and secure network communication.

Storage Virtualization

In traditional computing system, the storages have always been directly linked with the physical servers.

The virtualized storage systems are linked with servers and actual (physical) storage systems remain hidden.

virtualization of storage also happens through layer of software which creates logical abstraction of the pooling of physical storage devices having linked together by network.

Data stored in logical (virtualized) storage devices ultimately get stored in some physical storage disks. The advent of Storage Area Networks (SAN) has made the pooling (and hence the virtualization as well) of physical storage systems easier.

There are many commercial virtualized cloud storage systems available in the market. Google Cloud Storage, Microsoft's Azure Storage, Simple Storage System (S3) and Elastic BlockStore (EBS) of Amazon are few to name among them.

Desktop Virtualization

Desktop virtualization does not fall under the core category of computing infrastructure virtualization concept. But it is the key to business as it can lower the total cost of ownership and enhances security of system, application and data.

Desktop virtualization is different from remote desktop access. Through desktop virtualization technology, any computer's applications can be separated from its desktop and user can get the look and feel of some other environment while using those applications.

For instance, VMware Fusion 7 solution provides Mac-like experience while running Windows applications on a Mac system. In an enterprise environment, individual virtualized desktops can be maintained in a central server and users can access those desktops by connecting to the central server.

ADVANTAGES OF VIRTUALIZATION

Better Utilization of Existing Resources:

Physical computing resources have become advanced and powerful with time. In traditional computing, one machine instance runs on a physical server which hardly utilizes the whole power of the system.

Thus, most of the processing power simply remain unutilized for most of the computer systems. Running multiple virtual machines on one physical server makes better utilization of the resources and this is known as server consolidation.

Reduction in Hardware Cost:

As virtualization makes better use of physical resources by running multiple virtual machines on single set of physical resources, automatically cost of computing comes down.

If server consolidation can be combined with capacity planning in planned manner, investment for hardware resources may reduce drastically as well.

Reduction in Computing Infrastructure Costs:

Reduced physical computing resource requirements in turn reduces many other associated assets, like physical floor space, power requirement, cooling system and human resource to administrate the systems.

Increases in the number of virtual machines over existing physical resources do not add to any of these loads

Improved Fault Tolerance or Zero Downtime Maintenance:

The decoupling of virtual machines from specific hardware resources increases the portability of system.

In cases of any hardware failure, the virtual system can be migrated to another physical setup.

This helps to build fault tolerant system by creating scope for zero downtime maintenance.

Simplified System Administration:

Virtualization segments the management of the systems into two groups as physical resources management and virtual system management.

Centralized monitoring package can be employed to keep track of health of the systems and raise alert in case of need.

As managing virtual computing resources is less critical than physical resource and there are less physical machines, the system administration tasks become easier.

Simplified Capacity Expansion:

Capacities of virtual resources are easier to increase than expanding and then synchronizing physical computing resources. This also becomes possible due to the decoupling of physical resources from virtual systems.

Simplified System Installation: Installation of a new system has become easier, cost-effective and hassle-free in virtual environment.

A new system can be installed almost within no time by cloning a virtual machine instance. Fresh installation is much easier too than physical machine installation.

Support for Legacy Systems and Applications:

Research institutes or other organizations sometimes need to run legacy software packages which can no longer be run on presently available operating system or hardware platforms.

Virtualization is the only solution to tackle this kind of scenarios as any kind of VMs (including legacy systems) can easily be created to deploy required legacy applications over them.

Simplified System-Level Development:

System software (like device driver) development and testing require frequent re-booting of the system. This is easier and faster in virtual environment since VM rebooting does not require physical machine to restart and can be performed with some clicks of mouse.

Simplified System and Application Testing:

Performance testing of system software before its release is a rigorous job and requires to test the software on all supported platforms.

This is a difficult situation since it requires all of those hardware platforms and operating systems to test the software.

Virtualization eases this process by eliminating much of the time and effort required for system installation and configuration.

Application or system software testing is one of the biggest beneficiaries of virtualization.

Security:

Virtualization adds a layer of abstraction over physical hardware. Virtual machines cannot directly access physical resources any more.

This can restrict the amount of destruction that might occur when some malicious software attempts to damage the system or corrupt data.

For example, if an entire virtual hard disk gets damaged or corrupted, the actual physical disk remains unaffected.

DOWNSIDES OF VIRTUALIZATION

Virtualization is an enthralling technology which has changed the world of computing system design. But every technology has its own shortcomings and virtualization is no exception of this. Any weakness in virtualized system design especially in hypervisor may cause severe threat to the whole environment.

Single Point of Failure Problem:

The major benefit of virtualization is resource sharing.

Multiple virtual machines can run over one physical machine.

this has a downside.

It increases the probability of failure of a number of virtual servers in cases of failure of single physical machine.

Although, this situation can be handled easily by keeping backup resources and porting those virtual servers on the backup set of physical resources. Porting is not a difficult task as virtualization decouples virtual systems from physical resources.

Lower Performance Issue:

There is a concern whether virtual environments have the capacity to accomplish the full performance of the actual physical system.

It has been seen that virtual servers can achieve up to 85 percent to 90 percent of the performance of the actual physical server as VMs cannot get direct access to the hardware.

Difficulty in Root Cause Analysis:

With virtualization, a new layer of complexity is added which can cause new problems. The main difficulty is that if something does not work as it is supposed to it may require considerable extra efforts to find the cause of the problem.

VIRTUALIZATION SECURITY THREATS

The additional security threats to virtualized systems include the following:

The single point host:

Resource sharing makes virtualization cost-effective.

A single set of physical computing resources can fulfill multiple purposes and run several virtual machines.

Any security breach at this physical resource level may lead to a large number of system break-downs. So, the physical host level must be maintained at its best.

Threats to hypervisor:

Security and stability of any virtualized environment immensely depend on the ability of the hypervisor to protect itself from attacks.

Any security breach at the hypervisor level also makes the whole environment vulnerable.

Complex configuration:

Virtualization adds another layer of abstraction as well as complexity to computing systems.

These layers act as layer of protection to the system but at the same time any improper configuration greatly increases the probability of unseen vulnerabilities.

Privilege escalation:

Privilege escalation attack happens when users or some applications get access to more resources or functionalities than they are entitled to due to some design flaw in the system.

Like in any other software system, the privilege escalation problem may occur in virtualized system too where a hacker can attack a VM through the hypervisor.

Inactive virtual machines:

Virtual machines which are no more active or are in dormant state, generally moves out of the monitoring system automatically.

But, those machines may have sensitive data in storage or in their database.

As those remain out of the scope of the monitoring system it becomes a threat. Thus, loss of access to the inactive virtual machines causes potential security threats.

Consolidation of different trust zones:

Workloads of different trust levels from different zones consolidates onto same underlying physical system without adequate separation.

This may sometime create threat for the security of highly sensitive and trusted applications

VIRTUALIZATION SECURITY RECOMMENDATIONS

In addition to the traditional techniques some other security measures are necessary to be taken to ensure security of virtualized systems.

Hardening Virtual Machines:

In server virtualization, users have indirect access to computing resources through the virtual machines.

All applications they run or any computations they perform can be done on VMs only.

Robust and properly configured virtual machines will never allow any application to bypass them to directly access the hypervisor or underlying resources.

So, hardening the virtual machines should be a practice as they act as the first layer of defence.

The implementation may vary according to the vendor's recommendations. It is also important to keep virtual machine software updated to ensure that all known vulnerabilities have been corrected.

Hardening the Hypervisor:

Hypervisor is the key player in virtualization. Any communication between the virtual machines and the underlying resources are directed through the hypervisor.

So, it is inevitable to focus on the security of the hypervisor and make sure that the hypervisor is deployed steadily.

This ensures that even if any pitfall arises in any of the guest systems (the virtual machines), the VMM protects the other VMs and the underlying resources from any attack or further security breach.

Hardening the Host Operating System:

In hosted server virtualization technique, the host operating system plays vital role in managing the security of the physical system.

While any pitfall in the configuration of guest operating system can only affect the particular virtual machine's environment any pitfall in host OS may affect the entire environment as well as all of the guest machines.

Also a flawed host OS can weaken the hypervisor it is hosting to make the whole environment feeble.

Restrictive Physical Access to the Host:

Any vulnerability of the host system exposes an entire virtual environment to risks.

Host systems must be prevented from all external and unauthorized accesses. Any unauthorized physical access to the host system may easily make it vulnerable to attack in many ways.

Implementation of Single Primary Function Per VM:

Although the virtual machines are capable of handling multiple tasks, it makes the virtualization environment more secure if prime processes are separated among different VMs.

This isolation prevents the processes from being exposed and dilutes hacker's ability to damage multiple essential environmental functions when any weakness arises in one virtual machine.

Use of Secured Communications:

Establishing secured communication mechanisms provide protection to computing system. Encryption techniques should be used to frustrate hackers.

Techniques like Secure HTTP (HTTPS), encrypted virtual private networks (VPNs), transport layer security (TLS), secure shell (SSH) etc. help to prevent spoofed attacks and session hijacking.

Use of Separate NIC for Sensitive VM:

Virtual machines which process sensitive data will attract more attention from hackers over the network. In such scenario, it is better to use separate physical network interfaces card (NIC) for this type of virtual machines rather than sharing one NIC among multiple VMs.

Apart from the above listed measures, keeping all of the software, like host operating system, hypervisor, guest operating systems and others up-to-date and timely patching, disabling unnecessary services are also very important factors to enhance the security of a virtualized environment

VIRTUALIZATION AND CLOUD COMPUTING

Virtualization laid the foundation for cloud computing. According to IT research firm, Gartner, the virtualization is the key enabler of most of the fundamental attributes of cloud computing.

Those attributes of cloud are :

Shared service:

Resource sharing capability among multiple users has been redefined with the introduction of virtualization.

Users remain unaware about the actual physical resources and cannot occupy any specific resource unit while not doing any productive work.

Thus, cloud computing delivers shared service among consumers using resource virtualization.

Elasticity: With virtualized resources, the underlying capacity of actual resources can be easily altered to meet the varying demand of computation.

Thus, virtual machines and other virtual resources exhibit elasticity as well as flexibility.

Service orientation:

Cloud computing is built upon the philosophy of service-oriented architecture for application development.

The implementation of this architecture becomes easier when virtualization is in place.

Metered usage:

In cloud computing, the services are billed on usage basis. The accurate measurement of resource consumption has been possible due to use of virtualized resources.

RESOURCE POOLING

- The utility service model of cloud computing requires to maintain huge amount of all types of computing resources to provide different services to consumers.
- For this purpose, cloud service providers create pool of computing resources. Effective pooling or grouping of resources requires appropriate system designing and architectural planning.
- In traditional computing model, the silos are made with very little or no inter-connections. On the other hand in cloud computing, the consumers use well-connected pool of computing resources.
- They gain almost no knowledge or control over the locations from where physical resources are allotted to them. In its out-of-best scenario, the providers sometimes ask for choice of geographic location (country or continent) from where a consumer wants to get resources.

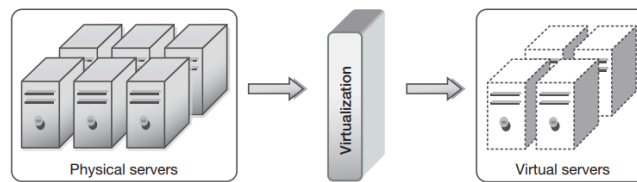


FIG 8.1: Pooling of physical and virtual servers

Resource Pooling Architecture

- A resource pooling architecture is designed to combine multiple pools of resources where each pool groups together with identical computing resources.
- The challenge is to build an automated system which will ensure that all of the pools get together in synchronized manner. Computing resources can broadly be divided into three categories as computer/server, network and storage.
- Hence, the physical computing resources to support these three purposes are essential to be configured in cloud data centers in good quantity. Again, a computer's capability mainly depends on two resource components like processor and memory.
- Thus, the resource pooling mainly concentrates on developing rich pools of four computing resources like processor, memory, network devices and storage.

1.1 Computer or Server Pool

- Server pools are developed by building physical machine pools installed with operating systems and necessary system software.
- Virtual machines are built on these physical servers and combine into virtual machine pool. Physical processor and memory components from respective pools are later linked with these virtual servers in virtualized modes to increase capacity of the servers.
- Dedicated processor pools are made of various capacity processors. Memory pools are also built in similar fashion. Processor and memory are allotted to the virtual machines as and when required.
- Again, they are returned to the pools of free components when load of virtual server decreases. Figure 8.2 shows a resource pool comprising of three separate pools of resources. This is a simplified demonstration of resource pooling. In reality, the pooling is implemented with other essential resources in more structured manner.

1.2 Storage Pool

Storage is another essential resource with rapidly growing requirement, frequently accessed by applications as well as with the consumers of computing. Storage pools are made of block-based storage disks. They are configured with proper portioning, formatting and are available to consumers in virtualized mode. Data having stored into those virtualized storage devices are actually saved in these pre-configured physical disks.



FIG 8.3: Storage pool

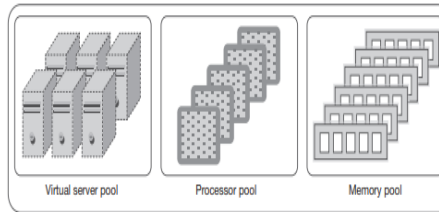


FIG 8.2: Resource pool comprising of three sub pools

1.3 Network Pool

Elements of all of the resource pools and the pools themselves owned by a service provider remain well-connected with each other.

This networking facilitates the cloud service at provider's end. As the cloud computing offers the facility of creating virtualized networks to consumers, the pool of physical network devices are also maintained in data centers.

Pools of networking components are composed of different preconfigured network connectivity devices like switches, routers and others. Consumers are offered virtualized versions of these components. They can configure those virtual network components in their own way to build their network.



FIG 8.4: Network pool

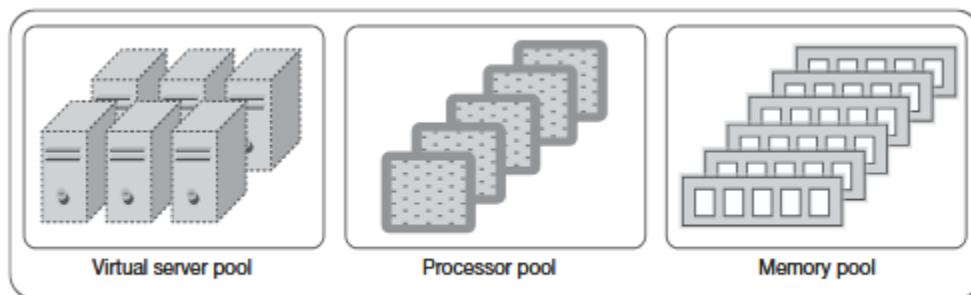


FIG 8.2: Resource pool comprising of three sub pools

1.3 Hierarchical Organization

- As stated earlier, the resource pool architecture generally maintains dedicated pools for different types of computing resources. Thus, data centers usually develop separate resource pools for server, processor, memory, storage and network components respectively.
- A self sufficient resource pool thus comprises of a number of smaller pools. With a very large number of resource components, a resource pool may become very complex to manage if not organized properly. Generally hierarchical structures are established to form parent and child relationships among pools.
- Figure 8.5 shows a sample hierarchical organization of pools where pools A1 and A2 are comprised of similar type of sub pools.
- When combined, they constitute a large pool. With such an organization, the pools eliminate the possibility of any single point of failure as there are multiple similar resource components which can handle load in case of failure of any component. Pools A1 and A2 may even be located at different geographic locations.

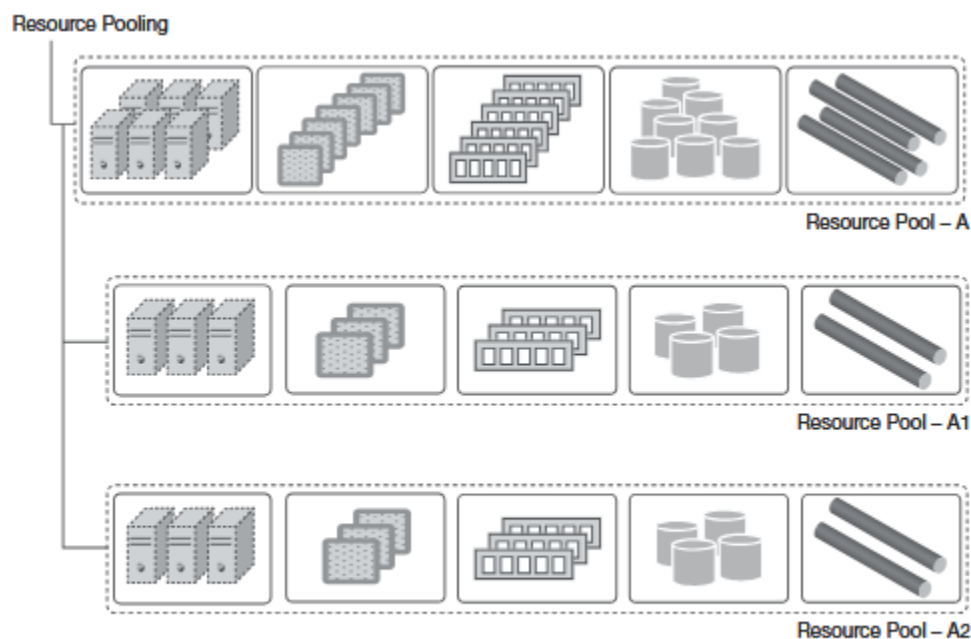


FIG 8.5: A sample resource pool organized in hierarchical manner. Each pool is comprised of five sub pools of server, processor, memory, storage and networking components.

COMMODITIZATION OF THE DATA CENTER

commodity hardware is a device component that is widely available, relatively inexpensive and more or less inter-changeable with other hardware of similar type. Cloud computing service promises to offer

high-performance computing (HPC) facility. Technologists have been succeeded to produce this high computing performance by combining the powers of multiple commodity computing components.

Commodity hardware components are those which are inexpensive, replaceable with similar component produced by other vendors and easily available.

Commoditization of resource pools at data centers has been an attracting feature of cloud computing, especially for the data center owners.

Specialized components are not required and commodity components are used to build the pools of (high performance) servers, processors, storage disks and else.

As commodity components are cheaper than specialized components, the commoditization of data centers is an important factor towards building of widely acceptable cloud computing facilities.

Unlike commodity computing components, the specialized components may not be easily available everywhere and any time.

The commoditization of data centers are also inspired by the diverse physical location of data centers on which clouds operate.

It is not always possible to keep the whole data center at one part of the globe for many reasons. Apart from location- wise benefits or disaster recovery issues, it is also a fact that many governments regulate the placement of physical storage of data within the jurisdiction of the business.

When commodity hardware components are used to build such data centers, it becomes easier for computing vendors to maintain the resource pools while it requires to replace and procure further.

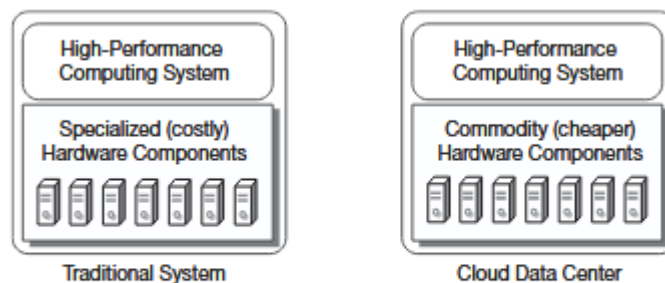


FIG 8.6: Traditional system versus commoditization of cloud data center

STANDARDIZATION, AUTOMATION AND OPTIMIZATION

All of the resource pools made of commodity hardware components are wrapped with virtualization.

Virtualization techniques of different resources (server, storage and network) represent corresponding set of methodologies on which common practices are developed.

Thus, virtualization layers provide opportunities to standardize, automate and optimize deployments of cloud environment.

Standardization:

Commodity resource components of a pool may come with various architectural standards.

Resource virtualization decouples the application instances from underlying hardware systems.

This creates the scope for simple and standardized logical representation of resources across the whole cloud environment.

Standardization is done based on set of common criteria.

Automation:

Automation is implemented for resource deployment like VM instantiation is used to bring VMs off-line back online and to remove them rapidly and automatically based on their previously set standard criteria.

Thus, the automation executes operations based on criteria set during the standardization process.

Optimization: This is the process which optimizes the resource usage. Optimization is performed to get optimal resource performance with limited set of resources.

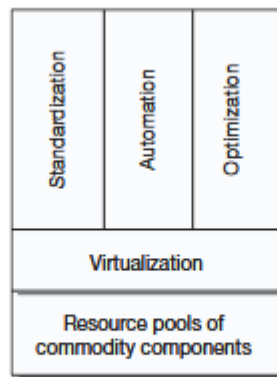


FIG 8.7: Standardization, automation and optimization at data center

RESOURCE SHARING

- Resource sharing leads to higher resource utilization rate in cloud computing. As a large number of applications run over a pool of resources, the average utilization of each resource component can be increased by sharing them among different applications since all of the applications do not generally attain their peak demands at same time.

- Cloud computing allows sharing of pooled and virtualized resources among applications, users and servers.
- The implementation needs appropriate architectural support. While servers are shared among users and applications the resources like storage, I/O and communication bandwidth are generally shared among several virtual machines.
- Resource sharing in utility service environment does not come without its own set of challenges.
- The main challenge is to guarantee the Quality of Service (QoS), as performance isolation is a crucial condition for QoS.
- The sharing may affect the run-time behaviour of other applications as multiple applications compete for the same set of resources. It may also be difficult to predict the response and turnaround time.

Multi-tenancy

- The characteristic of a system that enables a resource component to serve different consumers (tenants), where by each of them being isolated from the other is referred to as multi-tenancy.
- This is a feature available in public cloud model in its true and complete sense.
- Thus, any discussion about multi-tenancy in cloud computing primarily refers to public cloud deployment by default.
- In multi-tenancy model, the users cannot pre-occupy any particular physical resource component. Resource is utilized by different users on temporary basis as and when required.
- Multi-tenancy enables the service provider to provide computing service to multiple isolated consumers (who are not associated with each other) using single and shared set of resources.
- Multi-tenancy in cloud computing has been materialized based on the concepts of ownership- free resource sharing in virtualized mode and temporary allocation of resources from resource pool .
- It enables to lower computing costs and increases resource utilization which
- are important aspects of utility computing.
- The pooled resource components at cloud provider's end are used to serve a large number
- of service consumers.
- Cloud computing does not allow permanent acquisition of any resource component by any consumer. Whenever a consumer needs resources, appropriate resource components are delivered to the consumer in virtualized mode.
- The resources again get released whenever that consumer's needs are fulfilled and been returned to the pool of available or free resources which can be used to serve some other consumer's requirements in future.
- Thus, there is no one-to-one mapping between a consumer and a resource component in terms of resource use.
- Neither a resource component is dedicatedly used for some particular application. Resource components are allotted to users and applications solely on-availability basis.
- This increases resource utilization rate and in turn decreases investment as well.
- In cloud computing environment where a cloud service provider uses to serve a large number of consumers, this resource-sharing model through multi-tenancy has been capable to be changed into an effective mechanism for providing quality service at low cost.

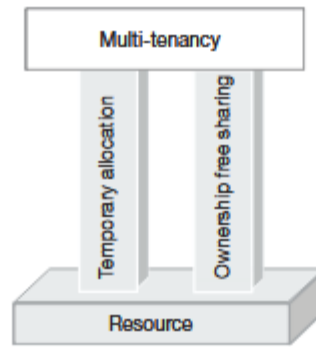


FIG 8.8: Philosophies behind the multi-tenancy

Types of Tenancy:

- Tenants of a system (multi-tenant system) can either be outsiders or even insiders within an organization (like multiple departments within an organization) where each of the tenants needs their own protected zones.
- In cloud computing whenever multi-tenancy is discussed, the focus shifts towards public cloud environment as the co-tenants being mutually exclusive or unknown to each other, share same computing infrastructure ideally without compromising their individual security and privacy.
- However, the idea of multi-tenancy also exists in community cloud and even in private cloud to some extent in alternative sense. Unlike the exclusiveness of the isolation and individuality of each tenant of a public cloud, the co-tenants in a community cloud are not mutually exclusive to each other as they belong to the same community and share similar interests.
- In private cloud, the concept of multi-tenancy is limited within the sub co-tenants internal to a single organization (or under a single tenant).
- The tenancy requirements of public and private clouds are almost similar irrespective of the fact that one has external co-tenants while the other has internal co-tenants.

Tenancy at Different Level of Cloud Services:

Similarly in cloud computing, the multi-tenancy is not limited at IaaS level, rather it covers PaaS and SaaS also.

At IaaS level, the multi-tenancy provides shared computing infrastructure resources like servers, storages etc.

At this level, multi-tenancy is achieved through virtualization of resources and different consumers share same set of resources through their virtual representation without affecting or being aware about one another.

At PaaS level, the multi-tenancy means sharing of operating system by multiple applications.

Applications from different vendors (tenants) can be run over same OS instance (to be offered as SaaS). This removes the need of allocating separate virtual machines per application by improving the maintenance and utilization of OS level capabilities.

At SaaS level, the multi-tenancy refers to the ability of a single application instance and/ or database instance (that is one database instance being created in some database servers) in order to serve multiple consumers.

Consumers (tenants) of SaaS share same application code base as well as database tables and codes (like procedure, functions etc.). Data of multiple consumers which are stored in same set of tables are differentiated by the respective consumer numbers.

Here, the separation of data is logical only although consumers get the feeling of physical separation. For application, the tenants may have the capability of customizing application functionalities or view of application interface but they cannot edit application code as the same application instance serves other consumers (tenants) also.

Multi-tenancy eases application maintenance (like updates) and makes it economical for provider as well as for the consumers.

In single-tenancy model, each consumer can be given the rights of the edit to application code as separate application instance is maintained for every consumers.

RESOURCE PROVISIONING

- In traditional computing, whenever a new server (or virtual server) is required to support certain workload, it takes lots of efforts and time of administrator to install and supply the server.
- Now a days computing needs rapid infrastructure provisioning facility to meet the varying demands of the consumers.
- With the emergence of virtualization technology and cloud computing IaaS model, it is now just a matter of minutes to achieve the same provided required volume of resources are being available.
- Thus, provisioning of a new virtual machine saves lots of time and effort in cloud.
- One can create a virtual server through a self-service interface which is considered as one of the most attractive feature of cloud.
- Flexible resource provisioning is a key requirement in cloud computing.
- To achieve this flexibility, it is essential to manage the available resources intelligently when required.
- Cloud provisioning is the allocation of cloud provider's resources to the consumers.

The Autonomic Way

Resource provisioning in cloud is the allocation of a cloud provider's resources directly to the consumers or to the applications.

When a consumer asks for resource, cloud provider must create appropriate number of virtual machines (VMs) in order to support the demand and should also allocate physical resources accordingly. This provisioning is an automated process in cloud which is designed by applying artificial intelligence and is known as autonomic resource provisioning.

The purpose of autonomic resource provisioning is to automate the allocation of resources so that the overall resource demand can be managed efficiently by minimum amount of resources. One resource is allotted to different applications or consumers.

This becomes possible since application loads varies with time. Through autonomic approach, computing Resource Pooling, Sharing and Provisioning resources can be rapidly provisioned and released with minimal management effort from a shared pool of configurable resources as on demand.

Role of SLA

Consumers typically enter into contract with cloud providers which describes the expected requirements of computing resource capacity being required for their applications. This contract is known as service level agreements (SLAs).

A cloud provider, after combining all such SLAs, can plan for the total amount of physical resources they have to keep as prepared to support all of its consumers so that those can be allocated to users when their applications would run.

This allocation is done dynamically by some provisioning algorithms that map virtual machines (VMs) running end-user applications into physical cloud infrastructure (compute nodes).

Cloud providers estimate the resource requirements of consumers through the SLA contract that consumers make with their providers.

Resource Provisioning Approaches

Efficient resource provisioning is a key requirement in cloud computing. Cloud consumers do not get direct access to physical computing resources.

The provisioning of resources to consumers is enabled through VM (virtual machine) provisioning. Physical resources from resource pool are made available to those VMs which in turn are made available to consumers as well as for the applications.

Physical resources can be assigned to the VMs using two types of provisioning approaches like static and dynamic.

In static approach,

VMs are created with specific volume of resources and the capacity of the VM does not change in its lifetime.

In dynamic approach, the resource capacity per VM can be adjusted dynamically to match work-load fluctuations. Computing resources in cloud computing are provisioned to consumers through virtual machine provisioning.

Static Approach

- Static provisioning is suitable for applications which have predictable and generally unchanging workload demands.
- In this approach, once a VM is created it is expected to run for long time without incurring any further resource allocation decision overhead on the system.
- Here, resource-allocation decision is taken only once and that too at the beginning when user's application starts running.
- Thus, this approach provides room for a little more time to take decision regarding resource allocation since that does not impact negatively on the performance of the system.
- Although static provisioning approach does not bring about any runtime overhead it has major limitations also.
- This provisioning approach fails to deal with un-anticipated changes in resource demands. When resource demand crosses the limit specified in SLA document it causes trouble for the consumers. Again from provider's point of view, some resources remain unutilized forever since provider arranges for sufficient volume of resources to avoid SLA violation. So this method has drawback from the viewpoint of both provider as well as for consumer.
- In static provisioning, resources are allocated only once during the creation of VMs. This leads to inefficient resource utilization and restricts the elasticity.

Dynamic Approach

With dynamic provisioning, the resources are allocated and de-allocated as per requirement during run-time.

This on-demand resource provisioning provides elasticity to the system.

Providers no more need to keep a certain volume of resources unutilized for each and every system separately, rather they maintain a common resource pool and allocate resources from that when it is required. Resources are removed from VMs when they are no more required and returned to the pool. With this dynamic approach, the processes of billing also become as pay-per-usage basis.

Dynamic provisioning technique is more appropriate for cloud computing where application's demand for resources is most likely to change or vary during the execution. But this provisioning approach needs the ability of integrating newly-acquired resources into the existing infrastructure. This gives provisioning elasticity to the system.

Dynamic provisioning allows system to adapt in changed conditions at the cost of bearing run-time resource allocation decision overhead.

This overhead leads some amount of delay in system but this can be minimized by putting upper limit on the complexity of provisioning algorithms.

Hybrid Approach

Dynamic provisioning addresses the problems of static approach, but introduces run-time overhead. To tackle with this problem, a hybrid provisioning approach is suggested that combines both static and dynamic provisioning.

It starts with static provisioning technique at the initial stage of VM creation and then turns it into dynamic re-provisioning of resources.

This approach can often effectively address real-time scenario with changing load in cloud computing.

Resource Under-provision and Over-provision Problems of Traditional Computing

Traditional computing systems mostly follow static resource provisioning approach. But, it is very difficult to correctly predict the future demand of any application (and hence the resource requirement for the application) despite rigorous planning and efforts.

This naturally results in under-provision or over-provision of resources in traditional environment.

When demand for computing resources crosses the limit of available resources, then a shortage of resource is created.

This scenario is known as under-provision of resource. A simple solution to this problem is to reserve sufficient volume of resources for an application so that resource shortage can never happen. But this introduces a new problem.

In such case, most of the resources will remain unutilized for majority of time. This scenario is known as over-provision of the resources. Under-provisioning problem occurs when the reserved resources are unable to fully meet the demand.

under-provision scenario.

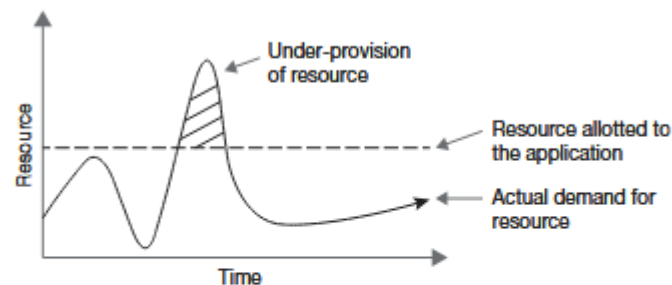


FIG 8.9: Resources under-provisioning problem in traditional system

Here, the allotted and defined volume of resource is represented by the dashed line. Under-provisioning problem occurs when resource demand of application is higher than this allotted volume. Under-provisioning causes application performance degradation.

The over-provisioning problem appears when the reserved volume of resource for an application never falls below the estimated highest-required amount of resource for the application considering the varying demand.

In such case, since for most of the time, the actual resource demand remains quite lesser than the reserved amount it ultimately turns into un-utilization of valuable resource. This not only causes wastage of resource but also increases the cost of computation.

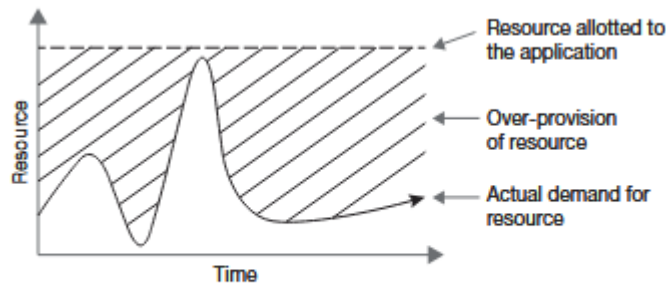


FIG 8.10: Resources over-provisioning problem in traditional systems

Resource over-provisioning problem occurs when the reserved resources remain unused for most of the time.

Figures 8.9 and 8.10 exhibit the problem of the traditional fixed-size resource allocation approach.

It tends to increase cost or provides poor performance outcomes. In cloud computing, the fine-tuned dynamic or hybrid resource provisioning approaches are used to deliver high performance at low cost.

While over-provisioning wastes costly resources, under-provisioning degrades application performance and causes business loss.

The static provisioning approach causes the trouble for vendors also. Vendors meet consumer's SLA requirements through resource over-provisioning in order to meet worst case demands.

However, since application demand remains low most of the time resource utilization rate also remains low. This restricts the vendors to deliver services at lower cost. Cloud computing addresses this issue by dynamic provisioning of resources using virtualization.

Resource Provisioning Plans in Cloud

Consumers can purchase cloud resources from the provider through web form by creating the account. Cloud providers generally offer two different resource provisioning plans or pricing models to fulfill consumers' requirements.

These two plans are made to serve different kind of business purposes. The plans are known as **short-term on-demand plan and long-term reservation plan**. Most commercial cloud providers offer both of the plans.

Short-Term On-Demand Plan

In this pricing model, resources are allotted on short-term basis as per demand. When demand rises, the resources are provisioned accordingly to meet the need. When demand decreases, the allotted resources are released from the application and returned to the free resource pool. Consumers are charged on pay-per-usage basis.

So, with this on-demand plan, the resource allocation follows the dynamic provisioning approach to fit the fluctuated and unpredictable demands.

In the short-term on-demand plan, it is the responsibility of the provider to ensure application performance by provisioning resources during high demand (the demand from multiple consumers at same time). This requires appropriate planning at provider's end. Estimation of resource requirements during high demand plays the vital role under this pricing model.

Application performance under short-term on-demand plan solely depends on provider's capability in estimating overall resource demand (of all consumers) during peak hours.

Long-Term Reservation Plan

In long-term reservation plan (also known as 'advance provisioning'), a service contract is made between the consumer and the provider regarding requirement of resources.

The provider then arranges in advance and keeps aside a certain volume of resources from the resource pool to support the consumer's needs in the time of urgency.

This arrangement of appropriating resources is done before starting of the service. In this model of resource provisioning, pricing is not on-demand basis.

Rather it is charged as a one-time fee for a fixed period of time generally counted in months or years.

At provider's end, the computational complexity as well as the cost is less under this plan in comparison to the on-demand plan.

This is because the provider becomes aware about the maximum resource requirement of the consumer and keeps the resource pool ready to supply resources to meet demands.

This reduces the provider's cost (of planning and implementation) and in effect the consumers may get the service in much cheaper rate (generally when long term contract is made) than the on-demand plan if hourly usage rates are considered.

The problem of the reservation plan is, since a fixed volume of resources is arranged as per the SLA there are possibilities of under-provisioning or over-provisioning of resources.

It is important for the cloud consumer to estimate the requirements carefully so that those problems can be avoided and at the same time the resource provisioning cost could minimize.

This goal can be achieved through an optimal resource management plan. Application performance under long-term reservation plan depends on consumer's ability to estimate their own resource demand in advance.

VM Sizing

VM sizing refers to the estimation of the amount of resources that should be allocated to a virtual machine.

The estimation is made depending on various parameters extracted out of requests from consumers. In static approach of resource provisioning, the VM size is determined at the beginning of the VM creation. But in dynamic approach of resource provisioning, the VM size changes with time depending on application load. The primary objective of VM sizing is to ensure that VM capacity always remains proportionate with the workload.

Proper VM sizing leads to optimal system performance.

VM sizing can be maintained in two different ways. Traditionally, VM sizing is done on a VM-by-VM basis which is known as **individual-VM based provisioning**. Here, resources are allotted to each virtual machine depending on the study of its previous workload patterns.

When additional resources are required for any VM to support load beyond its initial expectation, the resources are allotted from the common resource pool.

The other way is **joint-VM provisioning approach** where resources to VMs are provisioned in combined way so that the unused resources of one VM can be allocated to other VM(s) when they are hosted over

the same physical infrastructure. This approach takes advantage of the dynamic VM demand characteristics. Unutilized resources of less loaded VMs can be utilized in other VMs during their peak demand. This leads to overall resource capacity saving at provider's end. This technique is also known as VM multiplexing. Joint-VM provisioning saves significant volume from overall resource capacity compared to individual VM-based provisioning technique.

Dynamic Provisioning and Fault Tolerance

Dynamic resource provisioning brings many advantages to cloud computing over traditional computing approach. It allows runtime replacement of computing resources and helps to build reliable system.

This is done by constantly monitoring over all the nodes of a system executing some of the particular tasks. Whenever some of those nodes show low reliability over a predetermined period of time, a new node is introduced into the system to replace that defective or low performing node. This little effort turns the whole cloud system being more tolerant to faults. The reliability of nodes can be measured by monitoring various parameters like finding the number of faults occurred in near past.

Zero Downtime Architecture: Advantage of Dynamic Provisioning

The dynamic resource provisioning capability of cloud systems leads to an important goal of system design which is zero-downtime architecture.

One physical server generally facilitates or hosts multiple virtual servers. Hence, the physical server acts as the single point of failure for all of the virtual systems it creates. But the dynamic provisioning mechanism immediately replaces any crashing physical system with a new system instantly and thus the running virtual system gets a new physical host without halting.

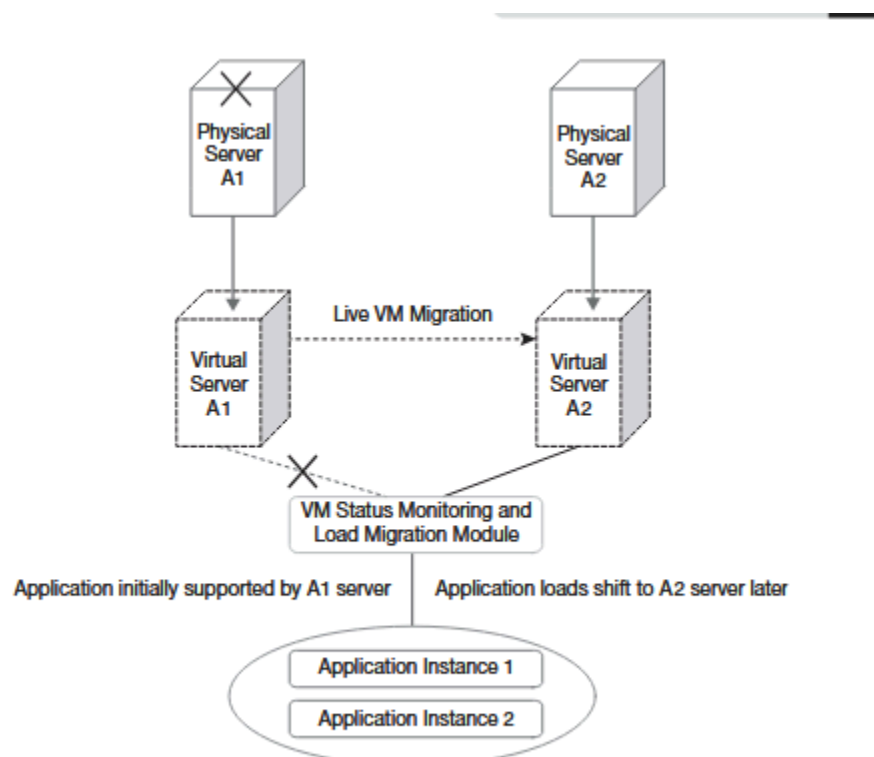


FIG 8.11: Live VM migration during failure of physical server

Figure 8.11 demonstrates how live VM migration maintains zero downtime during failure of physical host.

Here, the virtual server A1 (VM-A1) was hosted by physical server A1 and two applications were running on VM-A1. When physical server 'A1' crashes, VM-A1 is shifted to a new VM with current system status and all of the applications running in VM-A1 have been migrated to VM-A2 which is hosted by physical server 'A2'. Thus, the applications remain unaffected by the effects of zero downtime